Advanced computational methods for simulating the classical hydrogen atom in stochastic electrodynamics

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Stochastic electrodynamics (SED) is a theory of nature seeking to derive key aspects of quantum mechanics and quantum electrodynamics from a purely classical physics perspective. While SED has been successful in many areas, it has had mixed success when addressing the classical hydrogen atom bathed in classical electromagnetic zero-point radiation. Because analytical approaches are largely intractable, advanced numerical simulation methods are required. To address this, we develop a pseudo-wave formulation that integrates the solid angle in $k-$space utilizing symmetry, resulting in a significantly reduced memory footprint and lower computational overhead for generating the zero-point radiation spectrum. This optimization is paired with a logarithmic frequency grid, a sliding window tracking the instantaneous Keplerian frequency, and a Tukey smoothing function to eliminate discrete force discontinuities. Integrated via an adaptive RK4 algorithm with step-doubling across highly dense grids of up to 100-million bins, these techniques demonstrate the asymptotic convergence of ensemble stability. We show that increasing grid resolution in the high-frequency domain systematically mitigates artificial electron decay, yielding a simulated probability density function that closely aligns with the quantum mechanical ground state $P(r)$. Crucially, the simulations exhibit stable, low-eccentricity orbits without any observed atomic ionization events.

1.0. INTRODUCTION

This article pertains to a classical physics theory of nature usually referred to as stochastic electrodynamics (SED). Most work in SED has been carried out by applying analytical methods to linear and nonlinear oscillator systems. However, such approaches have not been successful when addressing physically realistic atomic systems. Consequently, computational attempts have been made in Refs. [1], [2], [3], [4]. These simulation studies have to do with the stochastic equilibrium of the classical hydrogen atom (H), where a classical electron is held bound to the nucleus by the Coulombic binding force (proportional to $1/r^2$), and the atom is "bathed" in electromagnetic (EM) zero-point (ZP) radiation. These two aspects of (1) a nonlinear binding force and the (2) stochastic interaction between charged particle and radiation, make the simulation problem difficult, particularly regarding problems of memory, speed, and accuracy. This article addresses these points. The importance of resolving these problems is no less than helping to learn whether SED may serve as a fundamental basis for quantum mechanics (QM).

The goal of the simulation program that will be described here is to determine whether a classical electron bound by the Coulombic force to the classical nucleus/proton, plus bathed in EM ZP radiation, will result in a classical H atom where the electron's orbit does not decay into the nucleus ("atomic collapse"), nor does the electron "ionize" from the nucleus. The program solves for the probability density $P(r)$ of the electron being at a distance r from the nucleus. If physically accurate, then $P(r)$ should agree with the predicted result from QM:

$$P_{\mathrm{QM}}\left(r\right)=\frac{4}{a_{\mathrm{B}}^{3}}r^{2}\exp\left(-\frac{2r}{a_{\mathrm{B}}}\right)\quad,\tag{1}$$

where a_{B} is the Bohr radius.

The three main solutions described here for improving the simulation capabilities in Refs. [1]-[4] are: (1) A means using symmetry in "$\vec{k}-$ space" to greatly reduce the number of random variables used in the computations, thereby greatly saving considerable memory and improving speed issues, (2) a logarithmic gridding method to improve speed and accuracy when dealing with radiation frequencies that span five to six orders of magnitude, and (3) several means of "smoothing" out the extremely rapidly varying stochastic force from the "zero-point" radiation, without sacrificing the essential interaction between radiation and charged particles. These approaches have not been used before in SED.

The first approach deals with the random variables that define the electric and magnetic field values at points in space and time. Using the electric dipole approximation (Sect. 3.0), it is shown that the random variables can be combined using symmetry of angular integration, thereby enormously cutting down on computational memory needs and improving speed. In terms of memory the improvement is from $O\left(N^{3}\right)$ to $O\left(N\right)$.

The second approach addresses the five to six orders of magnitude of the frequencies required of the electric and magnetic fields. Instead of a linear scaling of values, a logarithmic scaling is used (Sect. 4.1), which cuts down on electromagnetic field iterations at each point in time.

The third method addresses near infinite variations in time and space of the "zero-point" radiation in two ways: The first is to extract the main resonant part of the particle to radiation interaction (Sect. 4.2). Although this has been done before [1], the method is improved here by using the log scaling method just mentioned. The second part (Sects. 4.3-4.4) has to do with making sure that possible discontinuities in the ZP Lorentz force calculations are minimized.

The simulation studies in 2003 Ref. [1] and 2006 Ref. [2] deal with a "2D version" of the classical H atom. The present 2026 code agrees with the 2003/2006 work, but has now been developed for the full 3D situation. All three of these works differ considerably versus the 2015 3D H simulation work of Refs. [3] and [4]. References [3] and [4] predict rapidly changing eccentric orbits peaking as high as 0.8, and more importantly, that ionization occurs. This did not occur in the 2003/2006 codes, nor in the present code. Indeed, the eccentricity in these latter works remains very low, close to zero, which is that of nearly circular orbits. Likely the difference is the third point mentioned above of dealing with the seemingly discontinuous trajectories of the electron. No ionizations are predicted for the present code, nor for the 2003/2006 work. This will be described in more detail here in Sects. 4.2-4.4.

The "smoothing" of the Lorentz force is done here using a Tukey cosine function (Sect. 4.3). Together, all of these elements in Sects. 4.2-4.4.4 eliminate discrete step-changes in the simulated zero-point radiation, preventing the catastrophic numerical shocks that otherwise cause adaptive differential equation solvers to fail or predict spurious ionization-like effects. To a large extent these results are greatly aided with the first and second methods just mentioned to provide the necessary speed and memory improvements.

In addition, further techniques that have helped improve the simulation approach are discussed here, including

evaluation of the ZP Lorentz force at all integration points and a sliding window tracking the instantaneous Keplerian frequency. The trajectory integration of the electron is done in our C++ code via an adaptive Runge-Kutta fourth order (RK4) algorithm with "step-doubling" (also known as Richardson Extrapolation) [5] across highly dense grids of up to 100-million bins. These methods all combine to make the simulation code considerably faster and more accurate in its predictions.

The outline of this article is as follows: Sect. 2.1 presents a brief overview of SED for those not familiar with the theory, while Sect. 2.2 describes the typical starting point for mathematical analysis in SED. Section 2.3 discusses why the two main mathematical methods in Sects. 3.0 and 4.0 are important ones for aiding simulation computations in SED for the hydrogen atom and future atomic systems. Section 3.0 turns to the first of two main algorithm improvements, namely, a means using symmetry in "$k-$space" to greatly reduce the number of random variables used in the computations. Section 4.0 turns to a scaling means for working with frequency variables that span orders of magnitude in number, plus goes over a number of other computational improvements. Section 5.0 describes work carried out during the course of the development of the simulation software. This analysis helped confirm and direct the code/simulation creation to make sure the predicted physics and algorithms were working properly. Section 6.0 describes the computer code in more detail and shows examples of what can be obtained from the simulator. Section 7.0 ends with some concluding remarks. Ultimately, this article provides a comprehensive blueprint of the simulator's architecture, enabling researchers to leverage this publicly available software for advanced physical analysis in SED [6].

2.0. OVERVIEW OF SED

2.1. Background

We begin here with a brief overview of SED, including the variety of problems that have been successfully tackled, but also the difficulties encountered when moving onto more physically realistic atomic systems. The focus here will be on significant algorithm improvements that have made computer simulations of classical atomic hydrogen more feasible.

SED is a "classical" theory of nature, not in the sense that might initially be perceived, but rather in that it only involves the physics of the special relativistic version of Newton's second law of nature as applied to point charged particles (so the Lorentz-Dirac equation [7],[8]), and of Maxwell's equations. Hence SED consists of "classical" electrodynamics up to the approximate time of QM. It also assumes that classical EM radiation can exist at temperature $T = 0$ and in fact is essential to having equilibrium (*i.e.*, stochastic equilibrium) between classical charged particles and classical EM radiation. This radiation at $T = 0$ is referred to as classical electromagnetic ZP radiation. These few points have resulted in predictions in physical phenomena that are normally thought to only be addressable by QM and quantum electrodynamics (QED). Indeed, some physicists, including this author, have felt that many QM and QED aspects may be viewed as derivable from SED in the sense that thermodynamics is derivable from statistical mechanics.

Several reviews of SED exist such as Refs. [9] (SED up until 1996), [10], [2] (first half of article), and [11]. QM and QED phenomena that SED has been successful in addressing are: (1) Ref. [12] compares properties of the simple harmonic oscillator (SHO) within SED, QM, and QED; (2) van der Waals forces have been calculated to all distances and all temperatures [13], [14], [15]; (3) Casimir forces [9], [16]; (4) diamagnetism [17]; and (5) the thermal effects of acceleration [18], [19], [20], [21], [15], [22]. Yet more examples could certainly be cited.

Prior to the early 2000's the SED research largely involved applying various analytic methods to linear and nonlinear oscillator systems. However these analytic methods were not feasible for the nonlinear Coulombic binding force in atoms, combined with stochastic electromagnetic (EM) radiation bathing the system.

In 2003 a computer simulation method was developed to successfully move past this barrier [1],[2]. Very reasonable results were obtained as compared with quantum mechanics (QM). However, many approximations were made to enable feasible computations. In particular, one approximation involved making the atomic simulation be a "two dimensional (2D)" model. Fortunately in 2015 sufficient computer resources were garnered with improved algorithms to attack the problem in 3D [3],[4]. Computational issues have still persisted however. This article describes several key improvements in Sects. 3.0 and 4.0 to overcome these problems.

2.2. Mathematical start to SED

The computational framework of SED can be summed up via the following mathematical physics based descriptions of: (1) the classical electromagnetic fluctuating fields of radiation in space, and (2) the equations governing a system of classical charged point particles within this radiation. Regarding (2), the Lorentz-Dirac equation [7], [8], [23], [24] in special relativity is the key equation for the motion of these classical charged point particles, largely intended to represent electrons. Going beyond point particles is certainly of interest in SED. However, there has been less research pursued here as the charged point particles research has been difficult enough, and point particles are a natural description for electrons and a fairly good model for nuclei as well, at least at the atomic level.

The radiation fields analyzed here will be described by filling a large region of space, such that a classical model of an atomic system of nuclei and electrons in space are considerably smaller than the dimensions of this volume, and that the wavelength of the resonant frequency of the fluctuating electrons are similarly much smaller. At the end of our calculations here, the volume is usually extended to infinity. For simplicity of calculations, the volume of radiation is usually taken to be a rectangular parallelepiped region that the radiation fields are confined within. However, other shapes have been considered [25]. The rectangular parallelepiped considered here will have dimensions L_x, L_y, and L_z, along the x, y, and z axes, respectively.

The radiation fields will be represented as an infinite sum of plane waves, with periodic boundary conditions imposed. Thus we can then use Fourier series to describe the fields. Here we will use the following sum of plane waves for the electric $\vec{E}_{\rm rad}(\vec{x},t)$ and magnetic $\vec{B}_{\rm rad}(\vec{x},t)$ radiation fields in this large rectangular parallelepiped volume:

$$\vec{E}_{\rm rad}\left(\vec{x},t\right)=\frac{1}{\left(L_xL_yL_z\right)^{1/2}}\sum_{n_x,n_y,n_z=-\infty}^{\infty}\sum_{\lambda=1,2}\hat{\varepsilon}_{\vec{n},\lambda}\left[\begin{array}{c}A_{\vec{n},\lambda}\cos\left(\vec{k}_{\vec{n}}\cdot\vec{x}-\omega_{\vec{n}}t\right)\\+B_{\vec{n},\lambda}\sin\left(\vec{k}_{\vec{n}}\cdot\vec{x}-\omega_{\vec{n}}t\right)\end{array}\right]\ , \tag{2}$$

$$\vec{B}_{\rm rad}\left(\vec{x},t\right)=\frac{1}{\left(L_xL_yL_z\right)^{1/2}}\sum_{n_x,n_y,n_z=-\infty}^{\infty}\sum_{\lambda=1,2}\left(\vec{k}_{\vec{n}}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)\left[\begin{array}{c}A_{\vec{n},\lambda}\cos\left(\vec{k}_{\vec{n}}\cdot\vec{x}-\omega_{\vec{n}}t\right)\\+B_{\vec{n},\lambda}\sin\left(\vec{k}_{\vec{n}}\cdot\vec{x}-\omega_{\vec{n}}t\right)\end{array}\right]\ , \tag{3}$$

where

$$\vec{k}_{\vec{n}}=\frac{2\pi n_x}{L_x}\hat{x}+\frac{2\pi n_y}{L_y}\hat{y}+\frac{2\pi n_z}{L_z}\hat{z}\ \ , \tag{4}$$

n_x, n_y, and n_z are integers and $\vec{n}=n_x\hat{x}+n_y\hat{y}+n_z\hat{z}$ To satisfy all four of Maxwell's classical electrodynamic equations for fields in free space, then $\omega_{\vec{n}}=c\left|\vec{k}_{\vec{n}}\right|$, c is the speed of light, $\vec{k}_{\vec{n}}\cdot\hat{\varepsilon}_{\vec{n},\lambda}=\vec{k}_{\vec{n}}\cdot\hat{\varepsilon}_{\vec{n},\lambda'}=0$, and $\hat{\varepsilon}_{\vec{n},\lambda}\cdot\hat{\varepsilon}_{\vec{n},\lambda'}=0$ for $\lambda\neq\lambda'$, where λ and λ' indicate the linear polarization direction. Here, λ and λ' are indices that take on only two values, so each might be represented by the values 1 or 2, for example. Moreover, $\hat{k}_{\vec{n}}=\vec{k}_{\vec{n}}/\left|\vec{k}_{\vec{n}}\right|$; similarly, all other vectors with a "caret" are meant to be unit vectors.

It should be noted that one can show, using the free space Maxwell's equations, that Eqs. (2) and (3) satisfy the wave equations of $\nabla^2\vec{E}\left(\vec{x},t\right)=\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\vec{E}\left(\vec{x},t\right)$ and $\nabla^2\vec{B}\left(\vec{x},t\right)=\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\vec{B}\left(\vec{x},t\right)$. Moreover, the presence of $\hat{\varepsilon}_{\vec{n},\lambda}$ and $\left(\vec{k}_{\vec{n}}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)$ in Eqs. (2) and (3), plus the cited relationships of $\vec{k}_{\vec{n}}\cdot\hat{\varepsilon}_{\vec{n},\lambda}=\vec{k}_{\vec{n}}\cdot\hat{\varepsilon}_{\vec{n},\lambda'}=0$, and $\hat{\varepsilon}_{\vec{n},\lambda}\cdot\hat{\varepsilon}_{\vec{n},\lambda'}=0$ for $\lambda\neq\lambda'$, enable all four free space Maxwell's equations to be satisfied.

The stochastic nature of the radiation fields in Eqs. (2) and (3), as prescribed in SED, arises from the probability distribution of the Fourier coefficients $A_1,\cdots,A_N,B_1,\cdots,B_N$ over a large "ensemble" of equally sized space regions, each of which have the same dimensions of L_x,L_y,L_z. The radiation in the rectangular parallelepiped volumes in the ensemble are each characterized by different Fourier coefficients, resulting in a net probability distribution for each Fourier coefficient. It is in this way that the word "stochastic" enters into the name for SED. The radiation acts on charged particles, making the charged particles move differently in each member of the ensemble of radiation. The result is that the particles in the ensemble display stochastic properties as well. Each set of electrons and nuclei, plus the radiation fields, represents one member of the ensemble of moving electrons, nuclei, and fields.

In SED, the Fourier coefficients $A_1,\cdots,A_N,B_1,\cdots,B_N$ are assumed to be independent random variables with a Gaussian probability density distribution:

$$P_{\rm F}\left(A_p\right)=\frac{1}{\sqrt{2\pi\left(\sigma_p\right)^2}}\exp\left[-\frac{A_p^2}{2\left(\sigma_p\right)^2}\right]\ \ , \tag{5}$$

where the "p" index of $p=1,2,3,...$ is a shorthand notation for all the indices of $\vec{k}_{\vec{n}},\lambda$ in, for example, $A_{\vec{n},\lambda}$. The same probability density distribution in Eq. (5) also holds for the Fourier coefficient B_p. The label "F" is added to specify that this probability density function $P_{\rm F}$ refers to the Fourier coefficients. Moreover, σ_p depends on $\left|\vec{k}_{\vec{n}}\right|=\omega_{\vec{n}}/c$ and therefore on $|\vec{n}|$ as in Eq. (4). The functional form of σ_p at $T=0$ is a cornerstone for SED and expressed by

$$\sigma^2\left(\omega_{\vec{n}},T=0\right)=2\pi\hbar\omega_{\vec{n}}\ \ , \tag{6}$$

where σ_p^2 in Eq. (5) means this quantity in Eq. (6), and $\hbar$ is Planck's reduced constant. It is this simple linear function of ω at $T = 0$ that provides the Lorentz invariance property deduced by Marshall [26] and Boyer [27], and much later in Refs. [28], [29], [30] that Eq. (6) results in no heat flowing during isothermal, reversible processes . Imposing either of these properties onto Eqs. (2) and (3) enables the linear function of ω in Eq. (6) to be deduced, where the $2\pi\hbar$ factor arises by comparing with measurements of various physical phenomena.

When $T > 0$, the main spectrum of interest in SED is the "zero-point plus Planckian (ZPP)" radiation spectrum,

$$\sigma^2\left(\omega_{\vec{n}}, T\right) = 2\pi\hbar\omega_{\vec{n}} + \frac{4\pi\hbar\omega_{\vec{n}}}{\exp\left(\frac{\hbar\omega_{\vec{n}}}{k_{\rm B}T}\right) - 1} = 2\pi\hbar\omega_{\vec{n}} \coth\left(\frac{\hbar\omega_{\vec{n}}}{2k_{\rm B}T}\right) \quad , \tag{7}$$

where $k_{\rm B}$ is Boltzmann's constant. When $T = 0$, for the ZP case, this reduces to Eq.(6), $\sigma_p^2 = 2\pi\hbar\omega_{\vec{n}}$. For Rayleigh-Jeans (RJ) radiation, $\sigma_p^2 = 4\pi k_{\rm B}T$, which is what the ZPP expression in Eq. (7) becomes when $k_{\rm B}T \gg \hbar\omega_{\vec{n}}$.

Important properties of the A and B discrete variables are that

$$\langle A_{\vec{n},\lambda}\rangle = \langle B_{\vec{n}}, \lambda\rangle \quad \text{for all } \vec{n} \ . \tag{8}$$

$$\left\langle A_{\vec{n},\lambda} B_{\vec{n}',\lambda'}\right\rangle = 0 \text{ for any values of } \vec{n},\ \vec{n}',\ \lambda, \text{ and } \lambda'. \tag{9}$$

$$\left\langle A_{\vec{n},\lambda} A_{\vec{n}',\lambda'}\right\rangle = \delta_{\lambda\lambda'}\delta_{\vec{n}\vec{n}'} 2\pi\hbar\omega_{\vec{n}} \quad , \tag{10}$$

The $\langle\cdots\rangle$ brackets mean to take an ensemble average over all the members (radiation ZP field plus electron trajectory) of the ensemble. For each member of this ensemble, the $A's$ and $B's$ values are fixed. However, they are "randomly" different from one ensemble member to another, but with Eq. (5) describing their distribution of values in the net ensemble. Much of the above three equations is already embedded in the statement accompanying Eq. (5) that, "In SED, the Fourier coefficients $A_1, \cdots, A_N, B_1, \cdots, B_N$ are assumed to be **independent** random variables ... ."

The probability density of finding the fluctuating charged particle at position $\vec{x}$ in the steady state condition is [31]

$$\begin{aligned} P_{3\rm x}\left(\vec{x}\right) = & \int_{-\infty}^{\infty} dA_1 \cdots \int_{-\infty}^{\infty} dA_N \cdots \int_{-\infty}^{\infty} dB_1 \cdots \int_{-\infty}^{\infty} dB_N \\ & \times P_{\rm F}\left(A_1\right) \cdots P_{\rm F}\left(A_N\right) P_{\rm F}\left(B_1\right) \cdots P_{\rm F}\left(B_N\right) \delta^3\left[\vec{x} - \vec{x}_{A_1,\cdots,B_N}\left(t\right)\right] \quad , \end{aligned} \tag{11}$$

where $\delta^3\left[\vec{x} - \vec{x}_{A_1,\cdots,B_N}\left(t\right)\right]$ is the Dirac delta function in 3D space and the multiplication of all the $P_{\rm F}\left(A_i\right)$ and $P_{\rm F}\left(B_i\right)$ functions is due to $A_1, \cdots, A_N, B_1, \cdots, B_N$ being independent random variables. At the end of the calculations with Eq. (5), $N \to \infty$ is imposed. Finally, $\vec{x}_{A_1,\cdots,B_N\cdots}\left(t\right)$ in Eq. (11), gives the trajectory of the particle and is a function of $A_1, \cdots, A_N, B_1, \cdots, B_N$. All "$A$" and "$B$" coefficients are real quantities and are integrated from $-\infty$ to $+\infty$ in Eq. (11). Their values control the steady state solution for the particle's trajectory of $\vec{x}_{A_1,\cdots,B_N}\left(t\right)$. The probability density of these fields, $P_{\rm F}\left(A_1\right)...P_{\rm F}\left(A_N\right) P_{\rm F}\left(B_1\right)...P_{\rm F}\left(B_N\right)$, dictates their contribution regarding $\delta^3\left[\vec{x} - \vec{x}_{A_1,\cdots,B_N}\left(t\right)\right]$ by selecting the contribution to the final particle probability density $P_{3\rm x}\left(\vec{x}\right)$. More explanation on how to carry out such stochastic calculations involving the $A_{\vec{n},\lambda}$ and $B_{\vec{n},\lambda}$ coefficients is discussed in Refs. [31],[32],[33].

2.3. Why the computational methods aid in atomic solutions in SED

As discussed in Sect. I, computational methods now exist to solve problems of classical atoms that are represented not just by linear and nonlinear oscillators, but by more realistic atomic systems. These methods have so far only been applied to the simplest atom, namely, the classical hydrogen (H) system [1],[2],[3],[4]. Still, these methods look extendable to more complicated systems.

For just a single H atom, the frequencies of the electron orbiting the nucleus can range from the low frequency orbits of an electron in a Rydberg atom with large r, to the high frequencies of an electron in a close orbit to the nucleus. For the latter, the orbits and speeds would need to be described by special relativity (*i.e.*, the Lorentz-Dirac equation). The fluctuating motion of the orbiting electron dictates which frequencies of the EM radiation in Eqs. (2) and (3) are the most important ones to include. The frequency range of electromagnetic radiation emitted from an excited state of a hydrogen atom spans from about 2×10^9 Hz in the radio wave spectrum to about 3×10^{15} Hz in the UV spectrum, so 6 orders of magnitude. Sections 3.0 and 4.1 in this article address this point. Saving the values of $A_1, \cdots, A_N, B_1, \cdots, B_N$ over the entire frequency range of a simulation, introduces a severe problem that Sect. 3.0 deals with. Section 4.1 addresses the difficulty of summing over the related needed quantities at each time step, creating in particular a bottleneck at the high frequency end. Here the results of Sect. 3.0 are used to arrive at a better means in Sect. 4.1 for "indexing" the scale of the 10^6 range of variables. These computational methods reduce the memory storage and the amount of time retrieving and summing over the different parts of this information at each time step by many orders of magnitude.

3.0. WAVE-VECTOR SHELL INTEGRATION

3.1. General idea

The core algorithmic optimization in this model leverages spherical symmetry to integrate the 3D $k-$space field modes over all solid angles for a given frequency interval ω to $\omega + d\omega$. This procedure drastically reduces the number of random Gaussian variables that must be stored, retrieved, and summed at each integration time step, from scaling with ω^2 to ω^0 (*i.e.*, where ω^0 means no scaling with ω, but just dependent as usual on $\Delta\omega$).

The A and B variables in Eqs. (2) and (3) are generated as random numbers with a Gaussian probability density. The random variables A multiplies the $\cos(\omega t)$ term and the B random variable multiplies the $\sin(\omega t)$ term. As described here, by taking into account the vector nature of the polarization vectors, adding them all up and getting the net mean and variance, using symmetry, enables the net to be described by only 12 random numbers for each frequency interval of $\omega \to \omega + \Delta\omega$.

Moreover, in the large L limit, where $L_x = L_y = L_z = L$ in Eqs. (2) and (3), L cancels out of the analysis, where L is the length of a side of a large cube in 3D space. This makes perfect physical sense since the physical behavior of the electron should not depend on the periodically imposed value of L, which is just a mathematical means of dealing with the system. Moreover, although L_x, L_y, L_z need not be restricted in our analysis to the single value L, analyzing

a single classical hydrogen in free space makes this assumption of $L_x = L_y = L_z = L$ also physically reasonable. (For examples in SED where this restriction is not made, see Refs. [1] and [25].)

L drops out of the analysis as $L \to \infty$ due to two factors. The first is that the number of infinitesimal volume cells of size $\left(\frac{2\pi}{L}\right)^3$ within a large volume of a shell of k space is equal to

$$\frac{4\pi k^2 dk}{\left(\frac{2\pi}{L}\right)^3} .$$

This brings up the second point: Each small volume $\left(\frac{2\pi}{L}\right)^3$ that encloses a point of the k vector in k space, has an electric field amplitude A and B from Eqs. (2) and (3). Each amplitude is weighted by $1/\sqrt{V} = 1/L^{3/2}$, so the variance involving the square of these random Gaussian amplitudes is weighted by $1/L^3$ (the sum of the means is zero). The number of k points is weighted by L^3 from above, so the net dependence on L cancels out. This idea makes use of the fact that the sum of Gaussian random variables gives a Gaussian random number with the mean given by the sum of the means of the individual random numbers, and the net variance being the sum of the individual variances.

3.2. The statistical properties of the ZP electric field acting on the classical electron

The nonrelativistic ODE equation for a classical electron in a classical H atom is given by nucleus to the electron at time t,

$$m\frac{d^2}{dt^2}\vec{z}(t) = -\frac{e^2}{|\vec{z}|^3}\hat{z} + \vec{F}_{\rm rr}\left[\vec{z}(t),t\right] - e\vec{E}_{\rm ZP}\left[\vec{z}(t),t\right] - \frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\rm ZP}\left[\vec{z}(t),t\right] \quad , \tag{12}$$

where $\vec{z}(t)$ is the vector distance from the nucleus, m and $-e$ are the mass and charge of the classical electron, and

$$\vec{F}_{\rm rr}\left[\vec{z}(t),t\right] = m\tau\frac{d^3\vec{z}}{dt^3} \quad , \tag{13}$$

is the nonrelativistic radiation reaction (see for example Ref. [34]), where $\tau = \frac{2e^2}{3mc^3}$. In the case of H, the common approximation to Eq. (13) is:

$$m\tau\frac{d^3\vec{z}}{dt^3} = m\tau\frac{d}{dt}\left(\frac{d^2\vec{z}}{dt^2}\right) \approx m\tau\frac{d}{dt}\left(-\frac{e^2}{m\,|\vec{z}|^3}\hat{z}\right) \quad , \tag{14}$$

due to the strength of the binding force $-\frac{e^2}{|\vec{z}|^3}\hat{z}$, or assuming that

$$m\frac{d^2\vec{z}}{dt^2} \approx -\frac{e^2}{|\vec{z}|^2}\hat{z} \quad , \tag{15}$$

ignoring the contributions of $-e\vec{E}_{\rm ZP}\left[\vec{z}(t),t\right] - \frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\rm ZP}\left[\vec{z}(t),t\right]$. More will be said about this in Sec. 5.4.

One can show that Eq. (14) results in

$$\vec{F}_{\rm rr}\left[\vec{z}(t),t\right] = m\tau\frac{d^3\vec{z}}{dt^3} \approx \tau e^2\left[-\frac{\frac{d\vec{z}}{dt}}{|\vec{z}|^3} + \frac{3\vec{z}(\vec{z}\cdot\frac{d\vec{z}}{dt})}{|\vec{z}|^5}\right] \quad , \tag{16}$$

for the classical H case.

The probability/statistical aspects of the electron's trajectory originate from $\vec{E}_{\rm ZP}$ and $\vec{B}_{\rm ZP}$ and are passed onto the electron via Eq. (12). If we were treating the electron's trajectory via the full relativistic equation of motion (Lorentz-Dirac equation), the same basic idea would be true, meaning that the ZP fields's probability character will be transferred to the electron's trajectory in the same way. In a full analysis, the motion of the electron, with its accelerations and decelerations radiating energy, works in concert with the ZP field to influence its properties, particularly in a closed container. However, here we do not have a "closed container; the classical H atom is in empty space, but "bathed" in ZP field that then dictates the stochastic properties of the electron's trajectory.

We will apply the "dipole approximation" to help us obtain approximate solutions of Eq. (12). This step is a physical approximation that will help us create computer code to numerically obtain such solutions of Eq. (12) [23]. It consists of the following change of Eq. (12):

$$m\frac{d^2}{dt^2}\vec{z}\,(t) = -\frac{e^2}{|\vec{z}|^2}\hat{z} + \vec{F}_{\rm rr} - e\vec{E}_{\rm ZP}\,(0,t) - \frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\rm ZP}\,(0,t) \quad . \tag{17}$$

In Eq. (17), and from Eqs. (2) and (3), the dipole approximation means that

$$\vec{E}_{\rm ZP}\,(\vec{x},t) \approx \vec{E}_{\rm ZP}\,(0,t) = \frac{1}{(L)^{3/2}}\sum_{n_x,n_y,n_z=-\infty}^{\infty}\sum_{\lambda=1,2}\hat{\varepsilon}_{\vec{n},\lambda}\left[A_{\vec{n},\lambda}\cos\left(\omega_{\vec{n}}t\right) - B_{\vec{n},\lambda}\sin\left(\omega_{\vec{n}}t\right)\right] \quad , \tag{18}$$

$$\vec{B}_{\rm ZP}\,(\vec{x},t) \approx \vec{B}_{\rm ZP}\,(0,t) = \frac{1}{(L)^{3/2}}\sum_{n_x,n_y,n_z=-\infty}^{\infty}\sum_{\lambda=1,2}\left(\hat{k}_{\vec{n}}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)\left[A_{\vec{n},\lambda}\cos\left(\omega_{\vec{n}}t\right) - B_{\vec{n},\lambda}\sin\left(\omega_{\vec{n}}t\right)\right] \quad . \tag{19}$$

The dipole approximation is such that

$$\cos\left(\vec{k}\cdot\vec{x}-\omega t\right) \approx \cos\left(\omega t\right) \tag{20}$$

and

$$\sin\left(\vec{k}\cdot\vec{x}-\omega t\right) \approx -\sin\left(\omega t\right) \quad . \tag{21}$$

This is a good approximation when $\vec{k}\cdot\vec{x} \ll 1$. Here, $\left|\vec{k}\right| = \frac{2\pi}{\Lambda}$ where Λ is the wavelength of the light source being considered, is such that $\vec{k}\cdot\vec{x} < \left|\vec{k}\right||\vec{x}| = 2\pi\frac{|\vec{x}|}{\Lambda} \ll 1$. Another way, and perhaps a better way of applying these approximations that results in Eqs. (20) and (21), is when $\vec{k}\cdot\vec{x} \ll \omega t$, which certainly holds for small $|\vec{x}|$ and large t.

Considering first only the electric field in Eq. (18), we can imagine $\vec{k}_{\vec{n}}$ points in $k-$space as given by Eq. (4). The value of $A_{\vec{n},\lambda}$ and $B_{\vec{n},\lambda}$ is "indexed" by the two "labels" of $\vec{n}$ and λ. The probability density function in Eq. (5) describes the Gaussian distribution of values of $A_{\vec{n},\lambda}$ and $B_{\vec{n},\lambda}$ within the full ensemble of ZP radiation fields. But, just as $A_{\vec{n},\lambda}$ and $B_{\vec{n},\lambda}$ are Gaussian random variables, so are $\left(\hat{\varepsilon}_{\vec{n},\lambda}\right)_i A_{\vec{n},\lambda}$ and $\left(\hat{\varepsilon}_{\vec{n},\lambda}\right)_i B_{\vec{n},\lambda}$, where $\left(\hat{\varepsilon}_{\vec{n},\lambda}\right)_i$ for $i = 1,2,3$ are the x, y, z components of $\hat{\varepsilon}_{\vec{n},\lambda}$.

Suppose we consider all the "$\vec{n}$" points in k 3D space that lie within a shell volume of k space, described by $4\pi k^2 dk$. Rewriting Eq. (18):

$$E_{{\rm ZP},i}\,(0,t) = \frac{1}{(L)^{3/2}}\sum_{\substack{\text{All shells}\\ \text{of radius } k}}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\left(\hat{\varepsilon}_{\vec{n},\lambda}\right)_i\left[A_{\vec{n},\lambda}\cos\left(\omega_{\vec{n}}t\right) - B_{\vec{n},\lambda}\sin\left(\omega_{\vec{n}}t\right)\right] \quad . \tag{22}$$

In each "k shell", $\omega_{\vec{n}}$ is essentially constant for being within the infinitesimal dk of the shell of volume $4\pi k^2 dk$. Hence we can rewrite Eq. (22) as

$$E_{\mathrm{ZP},i}(0,t) = \sum_{\substack{\text{All shells}\\ \text{of radius } k}} \left\{ \begin{array}{l} \cos(\omega_{\vec{n}} t) \left[\frac{1}{(L)^{3/2}} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i A_{\vec{n},\lambda} \right] \\ -\sin(\omega_{\vec{n}} t) \left[\frac{1}{(L)^{3/2}} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i B_{\vec{n},\lambda} \right] \end{array} \right\} \quad . \tag{23}$$

Let

$$A_{\mathrm{E},i}(\omega) \equiv \frac{1}{(L)^{3/2}} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i A_{\vec{n},\lambda} \tag{24}$$

and

$$B_{\mathrm{E},i}(\omega) \equiv \frac{1}{(L)^{3/2}} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i B_{\vec{n},\lambda} \quad , \tag{25}$$

recognizing of course $\omega = ck$. $A_{\mathrm{E},i}(\omega)$ and $B_{\mathrm{E},i}(\omega)$ are Gaussian random variables, as they are the sum of Gaussian random variables. The mean of each is zero, since $\langle A_{\vec{n},\lambda} \rangle = \langle B_{\vec{n},\lambda} \rangle = 0$.

The variance is given by:

$$\begin{aligned} \langle A_{\mathrm{E},i}(\omega) A_{\mathrm{E},i}(\omega) \rangle \equiv \Sigma^2_{\mathrm{E},i}(\omega) &= \left\langle \left[\frac{1}{(L)^{3/2}} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i A_{\vec{n},\lambda} \right] \left[\frac{1}{(L)^{3/2}} \sum_{\vec{n}' \in \text{“}k \text{ shell”}} \sum_{\lambda'=1,2} (\hat{\varepsilon}_{\vec{n}',\lambda'})_i A_{\vec{n}',\lambda'} \right] \right\rangle \\ &= \frac{1}{L^3} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} \sum_{\vec{n}' \in \text{“}k \text{ shell”}} \sum_{\lambda'=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i (\hat{\varepsilon}_{\vec{n}',\lambda'})_i \langle A_{\vec{n},\lambda} A_{\vec{n}',\lambda'} \rangle \\ &= \frac{1}{L^3} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} \sum_{\vec{n}' \in \text{“}k \text{ shell”}} \sum_{\lambda'=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i (\hat{\varepsilon}_{\vec{n}',\lambda'})_i \delta_{\lambda\lambda'} \delta_{\vec{n}\vec{n}'} 2\pi\hbar\omega_{\vec{n}} \\ &= \frac{1}{L^3} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i^2 2\pi\hbar\omega_{\vec{n}} \quad . \end{aligned} \tag{26}$$

However,

$$\sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i^2 = 1 - \left(\frac{k_i}{\vec{k}} \right)^2 \tag{27}$$

Hence,

$$\begin{aligned} &\left\langle \left[\frac{1}{(L)^{3/2}} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \sum_{\lambda=1,2} (\hat{\varepsilon}_{\vec{n},\lambda})_i A_{\vec{n},\lambda} \right] \left[\frac{1}{(L)^{3/2}} \sum_{\vec{n}' \in \text{“}k \text{ shell”}} \sum_{\lambda'=1,2} (\hat{\varepsilon}_{\vec{n}',\lambda'})_i A_{\vec{n}',\lambda'} \right] \right\rangle \\ &= \frac{1}{L^3} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \left[1 - \left(\frac{k_i}{\vec{k}} \right)^2 \right] 2\pi\hbar\omega_{\vec{n}} \quad . \end{aligned} \tag{28}$$

Since

$$\omega_{\vec{n}} = \omega_{|\vec{n}|}$$

within this shell of constant k, then

$$\frac{1}{L^3} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \left[1 - \left(\frac{k_i}{\vec{k}} \right)^2 \right] 2\pi\hbar\omega_{\vec{n}} = \frac{2\pi\hbar\omega_{|\vec{n}|}}{L^3} \sum_{\vec{n} \in \text{“}k \text{ shell”}} \left[1 - \left(\frac{k_i}{\vec{k}} \right)^2 \right] \tag{29}$$

To evaluate the sum of points within the "k shell" of the quantity $\left[1-\left(\frac{k_i}{\vec{k}}\right)^2\right]$, the sum of any quantity over N points is of course exactly equal to the average value of that quantity multiplied by N. The question then becomes as to what is the average of $\left[1-\left(\frac{k_i}{\vec{k}}\right)^2\right]$ for points distributed uniformly around a spherical shell. Because of spherical symmetry, the average of $(k_x/k)^2$, $(k_y/k)^2$, and $(k_z/k)^2$ must all be exactly equal. Since they must add up to 1, the average value of each one is exactly 1/3. Therefore the average value of $\left[1-\left(\frac{k_i}{\vec{k}}\right)^2\right]$ is $\frac{2}{3}$. Hence:

$$\sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}} \left[1-\left(\frac{k_i}{\vec{k}}\right)^2\right] = \frac{2}{3}\times N_{\text{points.}}\ . \tag{30}$$

However, for $N_{\text{points.}} = (\text{number of points within } k \to k+dk)$, then

$$N_{\text{points}} = \frac{4\pi k^2 dk}{\left(\frac{2\pi}{L}\right)^3} = \frac{L^3}{2\pi^2}\frac{\omega^2 d\omega}{c^3}\quad . \tag{31}$$

Hence, the variance of $A_{\text{E},k,i}$ in Eq. (24) is

$$\Sigma^2_{\text{E},i}(\omega) = \langle A_{\text{E},i}(\omega)\, A_{\text{E},i}(\omega)\rangle = \frac{2\pi\hbar\omega}{L^3}\times\frac{2}{3}\times\frac{L^3}{2\pi^2}\frac{\omega^2 d\omega}{c^3} = \frac{2}{3\pi}\frac{\hbar\omega^3 d\omega}{c^3}\quad . \tag{32}$$

Thus, this variance is independent of L for large L, inversely proportional to c^3, and proportional to $\hbar$ and to ω^3. Also, $\Sigma^2_{\text{E},i}(\omega)$ has the same value for $i=1,2,3$. Moreover, everything above that was done for the variable $A_{\text{E},i}(\omega)$ in Eq. (24) also applies to the variable $B_{\text{E},i}(\omega)$ in Eq. (25).

It should be noted that $\langle A_{\text{E},i}(\omega)\, A_{\text{E},j}(\omega)\rangle$ equals zero, for $i\neq j$, since in the course of deriving it, along the steps above, one obtains $\sum\limits_{\lambda=1,2}\left(\hat{\varepsilon}_{\vec{n},\lambda}\right)_i\left(\hat{\varepsilon}_{\vec{n},\lambda}\right)_j$. But, this equals $-\frac{k_i k_j}{k^2}$, for $i\neq j$. When averaged over the solid angle in $k-$space, this equals zero.

In summary, we can then rewrite Eq. (18) as the following:

$$E_{\text{ZP},i}(0,t) = \sum_{\text{All } j}\left[A_{\text{E},i}(\omega_j)\cos(\omega_j t) - B_{\text{E},i}(\omega_j)\sin(\omega_j t)\right]\quad , \tag{33}$$

where the sum over j is the sum over frequencies, or over the value of ω_j in the jth "shell" in $k-$space. $A_{\text{E},i}(\omega_j)$ and $B_{\text{E},i}(\omega_j)$ are Gaussian random variables with mean given by$\langle A_{\text{E},i}(\omega)\rangle = \langle B_{\text{E},i}(\omega)\rangle = 0$, and variance $\left\langle A^2_{\text{E},i}(\omega)\right\rangle = \left\langle B^2_{\text{E},i}(\omega)\right\rangle = \Sigma^2_{\text{E},i}(\omega) = \frac{2}{3\pi}\frac{\hbar\omega^3 d\omega}{c^3}$, independent of i. Each of the six random numbers $A_{\text{E},i}(\omega)$ and $B_{\text{E},i}(\omega)$, for $i=1,2,3$ are described by the probability density distribution

$$P(a) = \frac{1}{\sqrt{2\pi\Sigma^2_{\text{E},i}(\omega)}}\exp\left(-\frac{a^2}{2\cdot\Sigma^2_{\text{E},i}(\omega)}\right)\quad . \tag{34}$$

Thus, $A_{\text{E},i}(\omega)$ and $B_{\text{E},i}(\omega)$ in Eqs. (24) and (25) each represent three random numbers for $i=1,2,3$, each with a mean of zero and a variance of $\Sigma^2_{\text{E},i}(\omega) = \frac{2}{3\pi}\frac{\hbar\omega^3 d\omega}{c^3}$, for a total of six independent Gaussian random numbers.

3.3. The statistical properties of the ZP magnetic field acting on the classical electron

The ZP magnetic field in the dipole approximation appears in Eq. (17) as $-\frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\text{ZP}}(0,t)$. We shall analyze $\vec{B}_{\text{ZP}}(0,t)$ in the same way we did for $\vec{E}_{\text{ZP}}(0,t)$ in the previous subsection.

From Eq. (19)

$$B_{\mathrm{ZP},i}\left(0,t\right)=\frac{1}{\left(L\right)^{3/2}}\sum_{\substack{\text{All shells}\\ \text{of radius } k}}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i\left[A_{\vec{n},\lambda}\cos\left(\omega_{\vec{n}}t\right)-B_{\vec{n},\lambda}\sin\left(\omega_{\vec{n}}t\right)\right]\quad . \tag{35}$$

In each “k shell”, $\omega_{\vec{n}}$ is essentially constant, for being within the infinitesimal dk of the shell of volume $4\pi k^2 dk$. Hence we can rewrite Eq. (35) as

$$B_{\mathrm{ZP},i}\left(0,t\right)=\sum_{\substack{\text{All shells}\\ \text{of radius } k}}\left\{\begin{array}{l}\cos\left(\omega_{\vec{n}}t\right)\left[\frac{1}{\left(L\right)^{3/2}}\ \sum\limits_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum\limits_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i A_{\vec{n},\lambda}\right]\\ -\sin\left(\omega_{\vec{n}}t\right)\left[\frac{1}{\left(L\right)^{3/2}}\ \sum\limits_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum\limits_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i B_{\vec{n},\lambda}\right]\end{array}\right\}\quad . \tag{36}$$

Let

$$A_{\mathrm{B},i}\left(\omega\right)\equiv\frac{1}{\left(L\right)^{3/2}}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i A_{\vec{n},\lambda} \tag{37}$$

and

$$B_{\mathrm{B},i}\left(\omega\right)\equiv\frac{1}{\left(L\right)^{3/2}}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i B_{\vec{n},\lambda}\quad , \tag{38}$$

each being Gaussian random variables. The mean of each is again zero, since $\left\langle A_{\vec{n},\lambda}\right\rangle=\left\langle B_{\vec{n},\lambda}\right\rangle=0$. The variance of $A_{\mathrm{B},i}\left(\omega\right)$ is

$$\begin{aligned}&\left\langle\left[\frac{1}{\left(L\right)^{3/2}}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i A_{\vec{n},\lambda}\right]\left[\frac{1}{\left(L\right)^{3/2}}\ \sum_{\vec{n}'\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda'=1,2}\left(\hat{k}'\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i A_{\vec{n}',\lambda'}\right]\right\rangle\\ &=\frac{1}{L^3}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\ \sum_{\vec{n}'\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda'=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i\left(\hat{k}'\times\hat{\varepsilon}_{\vec{n}',\lambda'}\right)_i\left\langle A_{\vec{n},\lambda}A_{\vec{n}',\lambda'}\right\rangle\\ &=\frac{1}{L^3}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\ \sum_{\vec{n}'\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda'=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i\left(\hat{k}'\times\hat{\varepsilon}_{\vec{n}',\lambda'}\right)_i\delta_{\lambda\lambda'}\delta_{\vec{n}\vec{n}'}2\pi\hbar\omega_{\vec{n}}\\ &=\frac{1}{L^3}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i^2 2\pi\hbar\omega_{\vec{n}}\quad .\end{aligned} \tag{39}$$

However, one can prove that

$$\sum_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i^2=1-\left(\frac{k_i}{\vec{k}}\right)^2\quad . \tag{40}$$

Hence,

$$\begin{aligned}&\left\langle\left[\frac{1}{\left(L\right)^{3/2}}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i A_{\vec{n},\lambda}\right]\left[\frac{1}{\left(L\right)^{3/2}}\ \sum_{\vec{n}'\ \in\ \text{“}k\text{ shell”}}\ \sum_{\lambda'=1,2}\left(\hat{k}\times\hat{\varepsilon}_{\vec{n},\lambda}\right)_i A_{\vec{n}',\lambda'}\right]\right\rangle\\ &=\frac{1}{L^3}\ \sum_{\vec{n}\ \in\ \text{“}k\text{ shell”}}\left[1-\left(\frac{k_i}{\vec{k}}\right)^2\right]2\pi\hbar\omega_{\vec{n}}\quad .\end{aligned} \tag{41}$$

Following now exactly the same set of steps as in the electric field case of the previous section, from Eqs. (29) to (32), the variance in Eq. (39) can be written as

$$\Sigma^2_{\mathrm{B},i}\left(\omega\right)=\frac{2}{3}\frac{\hbar\omega^3 d\omega}{\pi c^3}\quad , \tag{42}$$

which is equal to $\Sigma^2_{\mathrm{E},i}(\omega)$ in Eq. (32). The same step above for the variables $A_{\mathrm{B},i}$ apply for the variables $B_{\mathrm{B},i}$.

We can now rewrite Eq. (19) as

$$B_{\mathrm{ZP},i}(0,t) = \sum_{\mathrm{All}\ j} \left[A_{\mathrm{B},i}(\omega_j)\cos(\omega_j t) - B_{\mathrm{B},i}(\omega_j)\sin(\omega_j t)\right] \quad , \tag{43}$$

where the sum over j is the sum over each frequency ω_j in the jth "shell" in $k-$space. As before, each of the six random numbers $A_{\mathrm{B},i}(\omega)$ and $B_{\mathrm{B},i}(\omega)$ depend only on frequency, have zero mean, and variance $\Sigma^2(\omega) \equiv \Sigma^2_{\mathrm{E},i}(\omega) = \Sigma^2_{\mathrm{B},i}(\omega) = \frac{2}{3\pi}\frac{\hbar\omega^3 d\omega}{c^3}$. Thus each of these six "magnetic field ZP" variables are described by the same probability density distribution as before, namely, Eq. (34).

3.4. Concluding remarks regarding the "wave-vector shell" analysis

The statistical distribution of the Lorentz force $-e\vec{E}_{\mathrm{ZP}}(0,t) - \frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\mathrm{ZP}}(0,t)$ due to the ZP electric and magnetic fields acting on the classical electron in Eq. (17), imparts its statistical properties onto the ensemble of electron trajectories via Eqs. (33) and (43). Each of 12 independent random variables, $A_{\mathrm{E},i}$, $B_{\mathrm{E},i}$, $A_{\mathrm{B},i}$, $B_{\mathrm{B},i}$, for $i = 1, 2, 3$, has a mean of zero and variance $\Sigma^2(\omega) = \frac{2}{3\pi}\frac{\hbar\omega^3 d\omega}{c^3}$. Clearly this is enormously different than needing to provide random numbers for each point in k-space to the A and B coefficients in Eqs. (2) and (3). Instead, now for each interval $d\omega$, only 12 random numbers need to be supplied. Of course, the smaller $d\omega$ is made, the more accurate the stochastic properties are produced. To agree with physics in nature, saving the random number for each $d\omega$ interval is necessary and results in maintaining the radiation fields throughout time in the simulation.

Quantifying the exact reduction in memory and execution time provided by this algorithm is rather complex, where here we are concerned with all the A and B variable in Eqs. (2) and (3), for $\omega_{\min} \approx 10^{14}\frac{\mathrm{rad}}{\mathrm{s}}$ to $\omega_{\max} \approx 10^{19}\frac{\mathrm{rad}}{\mathrm{s}}$ (see Sect. 4.1 for these frequency range estimates). However, a rough approximation of the improvement in data storage requirements is as follows. Without the $k-$space integration method described from here, memory concerns start with the volume for a sphere in $k-$space, $\frac{4\pi k^3}{3}$, where from Eq. (4),

$$\left|\vec{k}_{\vec{n}}\right| = \frac{2\pi}{L}\left(n_x^2 + n_y^2 + n_z^2\right)^{1/2} \approx \frac{2\pi}{L}N \ . \tag{44}$$

Here, N is the approximate number of the k variables along the radius in the sphere in $k-$space. The number of $k-$space points in this sphere, times four variables at each point, is

$$\#\ \text{variables in sphere} = 4\times\frac{\left(\frac{4}{3}\pi k^3\right)}{\left(\frac{2\pi}{L}\right)^3} = \frac{16}{3}\pi N^3 \quad . \tag{45}$$

In contrast, the number of variables to be stored in this space using the "pseudo wave" analysis presented here is approximately given by the following, where δN is the number of k points in each δk interval of points, where $\delta k = \frac{\delta\omega}{c}$:

$$\#\ \text{variables in}\ N\ \ k-\text{points} = 12\times N\times\frac{\delta k}{\delta N} \quad . \tag{46}$$

More precisely, $\delta\omega$ is the interval in Eq. (31). Thus the method presented here for the total number of variables to be stored scales as N, where the conventional way above varies as N^3.

This improvement, along with the log-scaling method to be discussed in the next section, enables all random variables in the simulations to be stored totally within the computer's random access memory (RAM), thereby enormously speeding up retrieval of data for wave integrations carried out at each data point (twelve times, as will be discussed later [5]) in the simulation. Having the full integrations done at each point helps to make the ZP radiation Lorentz force be a smooth function without discontinuities.

4.0. LOGARITHMIC SCALING AND OTHER NUMERICAL IMPROVEMENTS

4.1. Scaling and indexing issues

Earlier in Sect. 1.0, a range of data was mentioned from radio waves to UV light, spanning a scale of about 10^6 for **emitted light** of frequencies from excited states (in QM terminology) to the ground state in H atoms. In terms of the classical electron orbital frequencies, that the program actually simulates, this approximate range can be deduced in the following way. For nonrelativistic circular motion, the angular frequencies of orbits come from balancing the mass times the centripetal acceleration of the classical H atom to the Coulombic force with the nucleus (cgs units):

$$mr\omega^2 = \frac{e^2}{r^2} \quad , \tag{47}$$

giving

$$\omega = \sqrt{\frac{e^2}{mr^3}} \quad . \tag{48}$$

This computer program largely covers $r = 25\,\text{Å}$ (although could readily be extended to much larger r values, but this has been "flagged" for approximating "ionization") to $r \approx 0.02\,\text{Å}$ (v/c here is about 4% the speed of light, so largely in the nonrelativistic regime). This range of orbital radii corresponds to about $1.3 \times 10^{14}\,\frac{\text{radians}}{\text{s}}$ to $5.7 \times 10^{18}\,\frac{\text{radians}}{\text{s}}$, or representing a span of more than four decades in orbital frequency.

Using a linear relationship and assuming that the lower frequency range requires at least 1000 data points to represent each $100\,\frac{\text{radian}}{\text{s}}$, results in

$$8\ \frac{\text{bytes}}{\text{word}} \times \frac{12\ \text{words}}{\text{point}} \times \frac{1000\ \text{points}}{100\ \text{rad/s}} \times (\text{Linear range}) \approx 6 \times 10^{21}\ \text{bytes} \quad , \tag{49}$$

for the number of bytes needed to represent the data. This occurs because $5.7 \times 10^{18}\,\frac{\text{radians}}{\text{s}} - 1.3 \times 10^{14}\,\frac{\text{radians}}{\text{s}} \approx 5.7 \times 10^{18}\,\frac{\text{radians}}{\text{s}}$, which represents the approximate physical linear range of data. Clearly 10^{21}bytes is orders of magnitude beyond what any RAM can hold. The situation would be much worse if the 12 words/point were not made possible by the "pseudo wave" work in Sec. 3.0, which changed the $k-$space scaling of memory from $O\left(N^3\right)$ to $O\left(N\right)$ scaling. Nevertheless, it is still not enough to deal with the ramifications of Eq. (49). A linear array that is dense enough to represent data in the lower frequency range of $10^{14}\,\frac{\text{rad}}{\text{s}}$ at $\frac{1000\ \text{points}}{100\ \text{rad/s}}$, when extended to $10^{18}\,\frac{\text{rad}}{\text{s}}$, results in the huge number in Eq. (49).

By implementing a logarithmic grid, the physical grid spacing $\Delta\omega$ is made in proportion to ω. This changes the required array size to the number of decades needed to represent the data, rather than the absolute linear range of

the data. When getting to the examples of the use of our software in Sects. 5.0 and 6.0, this will become even more meaningful, since log gridding sizes will still be large, but still acceptable. In our most aggressive sizes to date, grids exceeding 10^8 frequency bins will be utilized and discussed. Even though this is large, it is quite acceptable with today's modern computer hardware. Two improvements that make this possible is the reduction in spatial modes in Sect. 3.0 and the frequency log gridding here in Sect. 4.1.

The logarithmic gridding works is constructed by defining ω_n, the frequency at the nth point of the division of the spectrum, such that

$$\omega_{n+1} = K\omega_n \quad , \tag{50}$$

where K is a constant slightly larger than one. Consequently,

$$\ln(\omega_{n+1}) = \ln(\omega_n) + \ln(K) \quad . \tag{51}$$

As the frequencies grow exponentially, their logarithms grow linearly. Plotting the frequencies on a logarithmic axis results in them being evenly spaced by a distance of $\ln(K)$, hence the name "logarithmic grid" [35].

From Eq. (50),

$$\omega_{n+1} = K^2\omega_{n-1} = \ldots = K^{n+1}\omega_0 \quad , \tag{52}$$

so that as n grows, so does ω_n, but slowly, such that

$$\frac{\Delta\omega_n}{\omega_n} = \frac{\omega_{n+1} - \omega_n}{\omega_n} = \frac{(K-1)\,\omega_n}{\omega_n} = (K-1) \quad . \tag{53}$$

Thus, $\Delta\omega/\omega$ is a constant. As ω grows via the many magnitudes of its range, $\Delta\omega$ also grows.

Interestingly enough though, with the advantages in Sec. 3.0 and here in 4.1, memory is no longer the concern. Instead, the central processing unit's (CPU) integration time becomes the main problem. The true penalty of the linear grid is that numerically computing the Lorentz force in Eq. (12) at each time step requires summing over a huge number of discrete frequency values for the radiation fields, times twelve per adaptive time step. Even with the drastically reduced number of variables that the pseudo wave method provides, a large bottleneck occurs from a purely linear grid in terms of the CPU's integration time.

This numerical sum of the radiation field's frequencies at each time step, times twelve, and over a $\pm 5\%$ window (more on this will be described in Sect. 4.2) must be carried out. At the low end of the spectrum of $1 \times 10^{14}\,\frac{\text{rad}}{\text{s}}$ the divisions $\Delta\omega$ must be small enough (for example 100 such intervals) to carry out this integration somewhat accurately. However, keeping this same division at the high end of the spectrum of $6 \times 10^{18}\,\frac{\text{radians}}{\text{s}}$ becomes approximately

$$\left(6 \times 10^{18}\frac{\text{radians}}{\text{s}}\right) \cdot \left(\frac{100}{1 \times 10^{14}\,\frac{\text{radians}}{\text{s}}}\right) = 6 \times 10^6 \quad , \tag{54}$$

integration elements times twelve per time step. Thus, time step integrations become comparatively very slow at the high end of the spectrum. The logarithmic grid described here solves this problem by "stretching" the bins as ω increases.

Moreover, suppose p is such that $0 < p < 1$, then if a "window" is made of the frequencies about ω of $\omega(1-p)$ to $\omega(1+p)$, as is used in the algorithm here (Sect. 4.2), then

$$\ln[\omega(1+p)] - \ln[\omega(1-p)] = \ln\left[\frac{\omega(1+p)}{\omega(1-p)}\right] = \ln\left(\frac{1+p}{1-p}\right) \quad , \tag{55}$$

which is a fixed quantity, only dependent on p and true for all ω in the spectrum.

Put qualitatively, the frequency spectrum has to span from at least $10^{14}\frac{\text{rad}}{\text{s}}$ (the slow outer orbits) all the way to nearly $10^{19}\frac{\text{rad}}{\text{s}}$. If one was to use a pure linear grid over that entire span, a devastating trade-off would occur: preserving the necessary $\Delta\omega$ resolution for low frequencies would yield an astronomical number of bins at the high-frequency regime, paralyzing the CPU at every time step. The logarithmic scaling method bypasses this trap by enforcing a constant $\Delta\omega/\omega$ ratio, concentrating computational resolution only where the physical gradients demand it.

4.2. Resonance approximation and window algorithm

Despite the algorithmic improvements of the pseudo waves and logarithmic grid, the task of accurately computing and simulating the physics of SED is still impossible without a physically motivated truncation of the frequency spectrum. As can be seen in Eqs. (2) and (3) for the ZP fields, an infinite sum over the frequencies is present. Moreover, the magnitudes of these terms essentially only increase with frequency, as the variances of $A_{\vec{n},\lambda}$ and $B_{\vec{n},\lambda}$ are proportional to ω, as seen in Eqs. (5) and (6). One might then suspect that the SED physics must be "wrong." Instead, this divergence in SED parallels the infinities famously tackled by Feynman, Schwinger, and Tomonaga in QED renormalization. Fortunately, for the orbital dynamics of classical hydrogen in SED, the physical solution is considerably simpler. Likely when examining the fluctuations within an orbit, where the notion of "Zitterbewegung" starts to apply, then higher level corrections will be necessary.

The orbital "solution," at least for the H atom, was first carried out in 2003 in Ref. [1] and is repeated here in the present work. The infinite sum over frequencies is greatly simplified by considering the resonances between the radiation and the charged particle motion. (A deeper study is presently underway, namely, to recognize that the resonance is not just a function of frequency, but of EM strength as well.) Note that it is critical to the present simulation program that "windows" of physical resonances do indeed exist between the infinite frequency spectrum ZP radiation and the orbital motion of the classical electron. If this were not true, the present simulation task would be considerably harder.

The method is this: by considering a classical electron in a near circular orbit in a classical H atom, where the orbit is decaying (spiraling) inward due to radiation reaction, and by having a monochromatic circularly polarized (CP) EM wave acting, then a strong resonance is observed when the orbit's frequency comes close to that of the CP EM wave. Figures 9(a) and 9(b) in Ref. [36] indicate how this was realized. The orbits of the classical electron in the H atom have a strong response to the frequencies of the radiation acting on it. At the time of our 2003 and 2004 work, a "window" of $\pm$ 3% of the semimajor axis a of the orbit, which results in a window of $\pm$4.5% in the net angular frequency, as in Eq. (48), was found to be satisfactory. In the simulations reported here, a window of $\pm$ 5% has been

used to ensure inclusiveness.

This resonance approach is what Sect. 4.1 was referring to when summing over the waves within a window [see near Eq. (55)]. This resonance approach means that when evaluating the Lorentz force of $-e\vec{E}_{\rm ZP}\left[\vec{z}\left(t\right),t\right]-\frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\rm ZP}\left[\vec{z}\left(t\right),t\right]$ in Eq. (12), at time t, only the frequencies of $\vec{E}_{\rm ZP}$ and $\vec{B}_{\rm ZP}$ within $\pm$ 5% of the orbital frequency need be included. This infinity of the ZP spectrum is thus dynamically reduced to a finite numerical summation.

To aid in carrying out this "sliding window" approach with the orbital frequency f, where $\omega=2\pi f$, a least squares fit was used to dynamically determine the semimajor axis length "a", the eccentricity ε, and the angle of tilt of the semimajor axis θ_0, in the following elliptical equation for the radius $r\left(t\right)$ and angle $\theta\left(t\right)$ of the Coulombic orbit:

$$r\left(t\right)=\frac{a\left(1-\varepsilon^2\right)}{1-\varepsilon\cos\left[\theta\left(t\right)-\theta_0\right]}\quad . \tag{56}$$

If the ZP radiation did not exist, nor radiation reaction, then Eq. (56) would be the correct elliptical orbit of the electron. With ZP radiation and radiation reaction present, however, then Eq. (56) is only approximately valid, but reasonably close to the actual orbit for a number of orbits, due to the strength of the H nucleus acting. Consequently, a, ε, and θ_0 are all slowly varying functions of time t. These parameters were obtained by rewriting Eq. (56) as

$$\frac{1}{r}=\frac{1}{a\left(1-\varepsilon^2\right)}-\left[\frac{\varepsilon\cos\theta_0}{a\left(1-\varepsilon^2\right)}\right]\cos\theta-\left[\frac{\varepsilon\sin\theta_0}{a\left(1-\varepsilon^2\right)}\right]\sin\theta\quad . \tag{57}$$

By making a table of $1/r$, $\cos\theta$, and $\sin\theta$, for N points of an orbit or orbits, the parameters $\frac{1}{a(1-\varepsilon^2)}$, $-\left[\frac{\varepsilon\cos\theta_0}{a(1-\varepsilon^2)}\right]$, and $-\left[\frac{\varepsilon\sin\theta_0}{a(1-\varepsilon^2)}\right]$ were obtained using the least squares method since these parameters appear linearly in the above relationship. From these three extracted parameters, a, ε, and θ_0 were obtained.

Upon finding a, then via Kepler's third law, one obtains the period T_0, and then the orbital frequency is deduced by $f=\frac{1}{T_0}$. This least squares fit method was developed and published in Ref. [37]. This method was used here to obtain a close track of the net (central) angular frequency $\omega_c=2\pi f$, and therefore on the sliding window frequencies to be used in the Lorentz force summation of the ZP fields. It is this summation over frequencies that is carried out at each integration point (12 times) in the orbits.

This last point brings us back to the use and advantages of the logarithmic gridding system in Sect. 4.1. The exact number of frequencies to be summed over in the $\pm5\%$ window depends on the total grid size $N_{\rm total}$, which means the total number of discrete frequencies in the overall simulation. Because of the logarithmic scale, the $\pm5\%$ window always contains a fixed fraction of the $N_{\rm total}$ array, roughly 0.94%. More specifically, the total spectrum spans from $\omega_{\rm min}=1.3\times10^{14}$ to $\omega_{\rm max}=5.7\times10^{18}\,\frac{\rm rad}{\rm s}$. $\ln\left(\frac{5.7\times10^{18}}{1.3\times10^{14}}\right)\approx10.688$. The sliding window spans from $0.95\omega_c$ to $1.05\omega_c$. $\ln\left(\frac{1.05}{0.95}\right)\approx0.1001$. In terms of the logarithmic gridding, the fraction of the number of frequencies inside the sliding window to the fraction of the net wave frequency span is

$$\frac{\ln\left(\frac{1.05}{0.95}\right)}{\ln\left(\frac{5.7\times10^{18}}{1.3\times10^{14}}\right)}\approx0.0094\quad .$$

Therefore, the $\pm5\%$ window always captures approximately 0.94% of the total grid, or the total number of discrete frequencies used in the frequency range from $\omega_{\rm min}$ to $\omega_{\rm max}$.

Putting some numbers on this, if there are 1M discrete frequencies (bins) in the simulation, the CPU performs the sum of 9,360 Lorentz force $\vec{F}_{\text{Lor}}$ values per time step, times three, since $\vec{F}_{\text{Lor}}$ is a vector in 3-space, and times 12 since the RK step-doubling method is used. If 10M net discrete frequencies are used, then 93,600 discrete frequencies of the Lorentz force is summed over, and with an aggressive 100M bins, the most tried to date, this balloons to 936,000 per time step, times the previous factors. Despite being large, the $\pm 5\%$ window restricts the CPU to computing less than 1% ($\approx$0.94%) of the total domain at any given integration step, which helps make the simulation computationally tractable.

4.3. Smoothing function and related discontinuity avoidances

The sliding window described in the previous section is based on the approximate resonance width of the nearly elliptical orbit interacting with the ZP radiation. However, applying a strict rectangular truncation—where frequencies within the interval $\omega_c\,(1-p)$ to $\omega_c\,(1+p)$ contribute strongly to the Lorentz force, while frequencies immediately outside contribute nothing—introduces an artificial discontinuity. Physically, the influence of the radiation spectrum on the electron's resonance cannot fall off instantaneously. That makes little sense. Moreover, computationally, a hard rectangular boundary guarantees that whenever a discrete wave enters or exits the sliding window during integration, the differential equation solver will encounter a severe, non-physical step-change in the calculated Lorentz force.

If the discontinuity is allowed for either reason, the effect will be the same. If a wave amplitude instantly drops from 1 to 0 at exactly the frequency "window" edge, the derivative of the Lorentz force will mathematically spike to infinity. The adaptive RK4 step-doubling algorithm will interpret this as a massive local truncation error; it will "halve" the time step until it hits the minimum time step set in the integration software, which either results in the software aborting or getting trapped in an inescapable bottleneck.

To resolve this, a smoothing function was introduced to render the frequency truncation mathematically continuous. The cosine-tapered Tukey window [38], [39] provides exactly this capability. By applying this function, the amplitudes of the contributing EM waves are smoothly "tapered" to zero at the edges of the frequency interval. This not only captures the physical reality of a continuous resonance drop-off, but also provides a vastly superior numerical capability for the integration algorithm. The "window" of the frequency modifies the wave's amplitude contribution by the following factor, as a function of ω:

$$W(\omega) = \begin{cases} \frac{1}{2}\left[1+\cos\left(\pi\left(\frac{\omega-\omega_1}{\Delta_{\text{tapper}}}-1\right)\right)\right] & \omega_{\min} \le \omega < \omega_1 \\ 1 & \omega_1 \le \omega \le \omega_2 \\ \frac{1}{2}\left[1+\cos\left(\pi\left(\frac{\omega-\omega_2}{\Delta_{\text{tapper}}}\right)\right)\right] & \omega_2 < \omega \le \omega_{\max} \\ 0 & \text{otherwise} \end{cases}, \tag{58}$$

where Δ_{taper} is the frequency width of the fade-in/fade-out regions, $\omega_1 = \omega_{\min} + \Delta_{\text{taper}}$ is where the flat top begins, and $\omega_2 = \omega_{\max} - \Delta_{\text{taper}}$ is where the flat top ends and the fade-out begins. Thus, the center of the above function provides

a flat 1.0 amplitude multiplier for $\omega_1 \leq \omega \leq \omega_2$, while the edges use a cosine curve to provide a smooth continuous transition to zero. The taper width can be numerically experimented with to achieve the best numerical stability. The control of this function in the present software will be clarified further in Sect. 5.0.

Another source of discontinuities and likely severe computational barrier is the following. As briefly mentioned in Sect. 4.2, a tempting computational shortcut is to evaluate the EM wave summation only once or a few times per orbit. Once the initial grid is established and all related physical parameters are stored (*i.e.,* $A's$ and $B's$ values in Sect. 3.0, 12 for each bin or $\Delta\omega$ frequency interval), the vast majority of the simulation's CPU time is spent carrying out these summations. Hence, only updating the sum every N^{th} integration step might seem a needed and reasonable computational approximation.

However, over such a "macroscopic" duration, the instantaneous phases and fields change too rapidly. Updating the sum so infrequently can introduce large discrete jumps in the computed Lorentz force, violently kicking the electron and generating non-physical spikes in the orbital trajectory. Such artificial numerical shocks will likely lead to false trajectory predictions, including premature orbital decays or spurious ionizations. Quite possibly the ionization events reported in Refs. [3], [4] were at least partially caused in this way. After all, the eccentricity of the orbit reported in Ref. [3] was highly spiked, with values peaking as high as 0.8. For an ellipse, $0 \leq \varepsilon < 1$, with $\varepsilon \approx 0$ being a circle. In contrast, most of our simulations to date have been yielding orbits close to being circular and not highly elliptic at all. *I.e.*, see panel #3 in Fig. 1. In hindsight, this result of nearly circular orbits seems physically plausible since ZP radiation is homogeneous and isotropic; consequently it "bombards" the electron from all sides, over and over again.

4.4. Need for Runge-Kutta adaptive time step algorithm

The RK4 algorithm with "step-doubling" (Richardson Extrapolation) serves as the primary numerical workhorse for the simulations presented here. While methods such as Bulirsch-Stoer or predictor-corrector algorithms can be highly efficient for certain systems, they struggle when evaluating the highly stochastic, rapidly fluctuating electromagnetic fields inherent to this model [5]. Bulirsch-Stoer, for example, relies on rational function extrapolation assuming a smooth underlying field, making it unsuitable for the "ragged" environment of the ZP radiation.

The step-doubling approach relies on the standard four-stage RK4 evaluation, but executes this three separate times (12 total evaluations) to compare different step sizes. Step-doubling is exceptionally robust for highly fluctuating trajectories because the two half-steps provide a highly accurate final state, but it is computationally heavier than the embedded Runge-Kutta-Fehlberg (RKF45) method.

As discussed in the previous subsection, discontinuous step-changes in the evaluated forces can introduce severe instabilities into the integrator's predictions—a vulnerability shared by all differential equation solvers. The implementation of the continuous smoothing functions detailed in Sect. 4.3 directly mitigates this, providing the RK4 solver with a continuously differentiable framework.

The adaptive step algorithm used here plays an important role when the size of the orbit shrinks or grows, by saving steps when not needed and vice-versa. It is essential when considering an elliptical orbit, particularly when the orbit

has a high eccentricity ε. The speed near the nucleus (periapsis) is then far larger than the speed of the electron at the extreme distance from the nucleus (apoapsis). It is not uncommon to require, for example, 20,000 integration points near the nucleus compared to 1,000 or fewer at the apoapsis.

To maintain accuracy across vastly different orbital regimes such as these, the adaptive step algorithm employs an error estimate. The algorithm adjusts the step size to meet this criterion. The result is that the adaptive time-step algorithm dynamically scales the integration increment to meet strict error tolerances without unnecessary computational overhead. With the computational demands of the core physics involving a rapidly varying Lorentz ZP force, both in time and space, this RK4 algorithm with step doubling is essential for carrying out the computational aspects detailed here.

5.0. DEVELOPMENT ANALYSIS AND VALIDATION

5.1. Overview

The development of a robust simulation for a classical electron in a hydrogenic potential, bathed in zero-point (ZP) electromagnetic radiation, presents a unique validation challenge: there are no exact intermediate analytical expressions for this specific system to benchmark against. The ultimate test of the code is solely whether the final SED predictions reproduce QM. Consequently, establishing rigorous intermediate validation checks during code development required leveraging related, solvable analytical systems.

This section goes over the primary checks used to confirm the physics and algorithms of the simulation engine. Subsection 5.1 compares the simulation's generated ZP fields against exact analytical results for the two-point spatial and temporal correlation functions. Subsection 5.2 benchmarks the simulation using the SHO by comparing the numerically generated variance of a charged particle's position against a known analytical SED expression. Finally, Subsections 5.3 and 5.4 provide a computational analysis justifying the inclusion or omission of specific terms within the equation of motion, Eq. (12). Together, these analyses confirmed the numerical stability and physical accuracy of the underlying architecture before applying it to the full hydrogen atom.

5.2. Comparison with two-point ZP field correlation functions

During the development of the 3D simulations, it was critical to verify that the discrete sum over the ZP modes using Gaussian-generated random amplitudes [see Eqs. (2), (3), (5)] correctly replicates the continuous physical vacuum. To test this, we evaluated several examples of the ensemble average of the two-point spatial-temporal ZP field correlation functions $\langle E_{\mathrm{ZP},i}(t_1,\vec{x}_1)E_{\mathrm{ZP},j}(t_2,\vec{x}_2)\rangle$ and $\langle E_{\mathrm{ZP},i}(t_1,\vec{x}_1)B_{\mathrm{ZP},j}(t_2,\vec{x}_2)\rangle$. One of these examples will be cited here, namely, of $\langle E_{\mathrm{ZP},x}(0,0)E_{\mathrm{ZP},x}(t,R\hat{y})\rangle$, to validate the stochastic field generation engine.

The code executes the following steps to ensure accuracy:

- Analytical Benchmarking: The program establishes two theoretical baselines. First, it calculates the exact closed-form analytical solution for an infinite spectrum from $\omega = 0$ to ∞. Second, it computes a finite-bandwidth

continuous integral (from $\omega_{\min}$ to $\omega_{\max}$) to account for the exact numerical truncation bounds used in the simulation.

- Logarithmic Grid Geometry (Precomputation): The engine initializes a logarithmic frequency grid composed of 10,000 discrete modes. For each frequency bin, it precalculates an isotropic random wave-vector direction ($\hat{k}$) and an orthogonal polarization plane, projecting this geometry to extract the polarization vector directions. It also establishes the zero-point standard deviation ($\sigma \propto \sqrt{\hbar\omega}$) for each individual mode.

- Monte Carlo Ensemble Generation: The simulation iterates over 10,000 distinct statistical universes (ensembles). For every frequency mode within each universe, it draws random Gaussian Fourier coefficients (A and B) governed by the precomputed σ variance.

- Field Evaluation and Density Scaling: In each ensemble, the code measures the simulated ZP fields at two locations: the spatial origin at $t = 0$ and $\vec{x} = 0$ and a target atomic distance R at time t. Critically, the amplitude of each mode is multiplied by a 3D density-of-states weighting factor. This specific mathematical scaling dynamically maps the 1D logarithmic grid back into the physical 3D continuum.

- Convergence and Comparison: The program multiplies the two ZP field values, accumulates them across the 10,000 ensembles, and calculates the final numerical average. This value is then output alongside the theoretical targets to demonstrate whether the randomized algorithmic approach accurately reproduces the exact analytical physics.

Separation R (A)	Analytical $\omega_{\max} \to \infty$	Analytical (Finite $\omega_{\max}$)	C++ Numerical Average (10k ensembles)
0.1	7.4544×10^{18}	2.0764×10^{8}	1.9351×10^{8}
1.0	7.4544×10^{14}	2.0732×10^{8}	2.0732×10^{8}
10.0	7.4544×10^{10}	1.7640×10^{8}	1.7297×10^{8}

Table 1: Three ways of evaluating $\langle E_{\mathrm{ZP},x}(0,0)E_{\mathrm{ZP},x}(t,R\hat{y})\rangle$. Columns 2 and 3 represent the conventional method with a fixed integrand format, evaluated to an infinite limit and a finite cutoff $k_{\max} = \omega_{\max}/c$, respectively.. Column 4 is evaluated as in this simulation program, where the integrand at each frequency bin is generated by a Gaussian random number according to Eq. (5). Columns 3 and 4 are compared, where the integrand is taken over a finite maximum $k_{\max} = \omega_{\max}/c$ value. The second, third, and fourth columns all have units of $\left(\frac{\mathrm{statvolt}}{\mathrm{cm}}\right)^2$.

From Ref. [22], Appendix C, the analytical form is given by:

$$\begin{aligned}
\langle E_{\mathrm{ZP},x}(0,0)E_{\mathrm{ZP},x}(t,R\hat{y})\rangle &= \frac{1}{2}\int d^3k \frac{\hbar\omega}{2\pi^2}\left(1-\frac{k_x^2}{k^2}\right)\cos\left(k_y R - kct\right) \\
&= \frac{\hbar c}{\pi}\int_0^{k_{\max}} k^3 dk \left[\frac{\sin(kR)}{kR} + \frac{\cos(kR)}{(kR)^2} - \frac{\sin(kR)}{(kR)^3}\right]\cos(kct) \\
&\underset{k_{\max}\to\infty}{\to} + \frac{4\hbar c}{\pi}\left\{\frac{R^2+(ct)^2}{\left[(ct)^2-R^2\right]^3}\right\} \quad . \qquad (59)
\end{aligned}$$

The second column in Table 1 is given by the third line of Eq. (59); the third column is given by the second line of Eq. (59), as evaluated in Mathematica. The fourth column is given by our simulator that sums up the random amplitudes at each frequency, weighted by the density of states, to construct the total electric field at the spatial origin at time $t = 0$. The simulator does the same thing at position $R\hat{y}$ and time t. Adding the products up in $E_{\mathrm{ZP},x}(0,0)E_{\mathrm{ZP},x}(t,R\hat{y})$, over 10,000 bins, yields column 4. As shown, the finite-bandwidth analytical results (Column 3) and the simulation output (Column 4) are in excellent agreement.

This validation procedure was successfully repeated for four different correlation functions, such as:

$$\langle E_{\mathrm{ZP},x}(0,0)B_{\mathrm{ZP},z}(t,R\hat{y})\rangle \underset{k\to\infty}{\to} -\frac{4\hbar c}{\pi}\left\{\frac{2Rct}{\left[(ct)^2-R^2\right]^3}\right\} \quad .$$

Similar agreement as in Table 1 were found for these different correlation functions.

5.3. Use of simulator to "randomly" evaluate $\langle r^2\rangle$ for the SHO

During the course of development, a further check was made on how well the summation over random generated modes, weighted by the ZP spectrum frequency functional value, agreed with an analytic expression. Instead of just a check on the fields $\vec{E}_{\mathrm{ZP}}$ and $\vec{B}_{\mathrm{ZP}}$, as in the previous subsection, here the effect of these fields was evaluated for the behavior of the classical electron in a SHO binding potential. The reason for choosing this case is that known analytic solutions existed for the SHO that could be used in making such a comparison. In particular, the ensemble average of the square of the radius $\langle r^2\rangle$ for a classical electron charge in a SHO binding potential can be evaluated analytically as well as by simulation.

The equation of motion examined was

$$m\frac{d^2}{dt^2}\vec{z}(t) = -m\omega_0^2\vec{z}(t) - m\tau\omega_0^2\frac{d}{dt}\vec{z}(t) - e\vec{E}_{\mathrm{zp}}(0,t) \quad . \tag{60}$$

This equation is a linear ordinary differential equation (ODE) that is also a stochastic one, because of the presence of the "driving term" $-e\vec{E}_{\mathrm{zp}}(0,t)$. It should be contrasted with the expression in Eqs. (17) and (13). In Eq. (60) above, $-m\omega_0^2\vec{z}(t)$ is the binding force for the electron in a SHO binding potential $\frac{m\omega_0^2\vec{z}^2}{2}$, with ω_0 being the resonant frequency of the oscillator. The radiation reaction $\vec{F}_{\mathrm{rr}}$ expression of $-m\tau\omega_0^2\frac{d}{dt}\vec{x}(t)$ came from approximating Eq. (13) via

$$m\tau\frac{d^3\vec{z}}{dt^3} \approx m\tau\frac{d}{dt}\left(-m\omega_0^2\vec{z}(t)\right) \quad , \tag{61}$$

following the same reasoning as in Eqs. (14) and (15).

The other major difference between Eq. (17) and Eq. (60) is that the term

$$-\frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\mathrm{ZP}}(0,t)$$

was dropped due to the assumption that

$$\frac{1}{c}\left|\frac{d\vec{z}}{dt}\right| \ll 1 \quad . \tag{62}$$

As discussed in many SED references such as Ref. [31] [see Eq. (36), times 3 because of three degrees of freedom]:

$$\left\langle r^2\left(t\right)\right\rangle \approx 3\frac{\left(e/m\right)^2}{\left(2\pi\right)^3}\frac{8\pi}{3}\frac{1}{c^3}\int_0^\infty d\omega\frac{\omega^2\sigma^2\left(\omega,T\right)}{\left(-\omega^2+\omega_0^2\right)^2+\left(\tau\omega^3\right)^2} \quad . \tag{63}$$

where $T=0$ for the ZP case. In the limit of $\tau\to 0$, and with $\sigma^2\left(\omega,T=0\right)=2\pi\hbar\omega$, this becomes

$$\left\langle r^2\left(t\right)\right\rangle \to \frac{3\hbar}{2m\omega_0} \quad , \tag{64}$$

a familiar expression for the expectation value of the variance of the 3D SHO position in QM. It should be noted that Eq. (63) was obtained by writing the steady state solution of $\vec{z}(t)$ in Eq. (60) as a function of $\vec{E}_{\rm ZP}\left(0,t\right)$ in Eq. (2), which in turn depends on the random variables of $A's$ and $B's$, and then taking the ensemble average $\langle\vec{z}\cdot\vec{z}\rangle$, using the properties of the $A's$ and $B's$ in Eqs. (8)-(10).

What was helpful about all of this in the development of the present simulation program is that Eq. (63) can be evaluated in two ways, one via analytically carrying out the integral via numerical means (such as via Mathematica), or by using the simulation program to numerically solve Eq. (60) over a length of time, assuming ergodicity, and finding the time average of

$$\left\langle r^2\left(t\right)\right\rangle \approx \frac{1}{T_{\rm m}}\int_0^{T_{\rm m}} dt\left|\vec{z}\left(t\right)\right|^2 \quad , \tag{65}$$

for large values of $T_{\rm m}$, and where the numerical values of $m=9.1094\times10^{-28}$g, $e=4.8032\times10^{-10}$esu, and $\omega_0=4.5\times10^{16}{\rm s}^{-1}$(frequency close to that of the Bohr orbit) were used. Instead of $\tau=\frac{2e^2}{3mc^3}=6.2660\times10^{-24}$ s, the values of $f\times6.2660\times10^{-24}$ s were inserted, where different values of f were used for testing and comparing Eqs. (63) and (65), as seen in the table below. For Eq. (65), $T_{\rm m}=1$ ps was used. As can be seen below, columns 2 and 3, representing the two methods, agreed quite well over this large range of values for f.

f	Mathematica result $\left(\text{cm}^2\right)$ of Eq. (63)	C++ Simulator result $\left(\text{cm}^2\right)$ of Eq. (65)
10^5	4.064×10^{-22}	3.814×10^{-22}
10^4	3.895×10^{-21}	3.824×10^{-21}
10^3	3.863×10^{-20}	3.968×10^{-20}

Table 2: Comparison of calculating $\left\langle r^2\left(t\right)\right\rangle$ via the analytic Eq. (63) and the simulated method of Eq. (65).

5.4. Check on ZP Lorentz force inclusion in the radiation reaction

The development of the simulation program described here included an examination of physical effects that should be included in Eq. (12). Two main areas in particular were explored: (1) examining including the contribution of $-e\vec{E}_{\rm ZP}\left[\vec{z}\left(t\right),t\right]-\frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\rm ZP}\left[\vec{z}\left(t\right),t\right]$ to the radiation reaction, and (2) the dipole approximation, which Sect. 3.0 (pseudo wave analysis) was partially based upon.

Regarding (1), the standard procedure in SED is to have the radiation reaction depend only on the Coulombic force, as in Eqs. (14)-(16). However, it is natural to ask about using

$$m\tau\frac{d^3\vec{z}}{dt^3}\approx\tau\frac{d}{dt}\left(-\frac{e^2}{\left|\vec{z}\right|^3}\hat{z}-e\vec{E}_{\rm ZP}\left[\vec{z}\left(t\right),t\right]-\frac{e}{c}\frac{d\vec{z}}{dt}\times\vec{B}_{\rm ZP}\left[\vec{z}\left(t\right),t\right]\right) \quad . \tag{66}$$

To examine this effect of the above extra terms in the radiation reaction expression, the dipole approximation was made in the fields $\vec{E}_{\rm ZP}$ and $\vec{B}_{\rm ZP}$ and the expressions of Eqs. (18) and (19) were used. However, the sum over all the modes in (18) and (19) was changed to include only the waves that were used in the equation of motion, for each value of ω, due to the "window resonance" discussed in Sect. 4.2.

Starting the simulation with the classical electron in a circular orbit of radius 0.53 Å, and using Eq. (66), with the restrictions just mentioned, the magnitude

$$\left| \tau \frac{d}{dt} \left(\frac{e^2 \hat{z}}{|\vec{z}|^3} \right) \right| \quad ,$$

which represents the Coulombic contribution to the radiation reaction, was plotted, as was the magnitude

$$\left| \tau \frac{d}{dt} \left(-e\vec{E}_{\rm ZP} \left[\vec{z}\left(t\right), t \right] - \frac{e}{c} \frac{d\vec{z}}{dt} \times \vec{B}_{\rm ZP} \left[\vec{z}\left(t\right), t \right] \right) \right| \quad ,$$

with the restrictions just mentioned, also included. Figure 1 shows the result, with the y-axis using a log scale to show the large difference in the magnitudes. As can be seen, the ZP contribution to $\vec{F}_{\rm rr}$ from the ZP Lorentz EM force is orders of magnitude smaller than the contribution of the Coulombic to $\vec{F}_{\rm rr}$. Hence, the approximation of

$$\vec{F}_{\rm rr} \approx \tau \frac{d}{dt} \left(\frac{e^2 \hat{z}}{|\vec{z}|^3} \right) = \tau e^2 \left[-\frac{\frac{d\vec{z}}{dt}}{|\vec{z}|^3} + \frac{3\vec{z}(\vec{z} \cdot \frac{d\vec{z}}{dt})}{|\vec{z}|^5} \right] \quad , \tag{67}$$

is an excellent one.

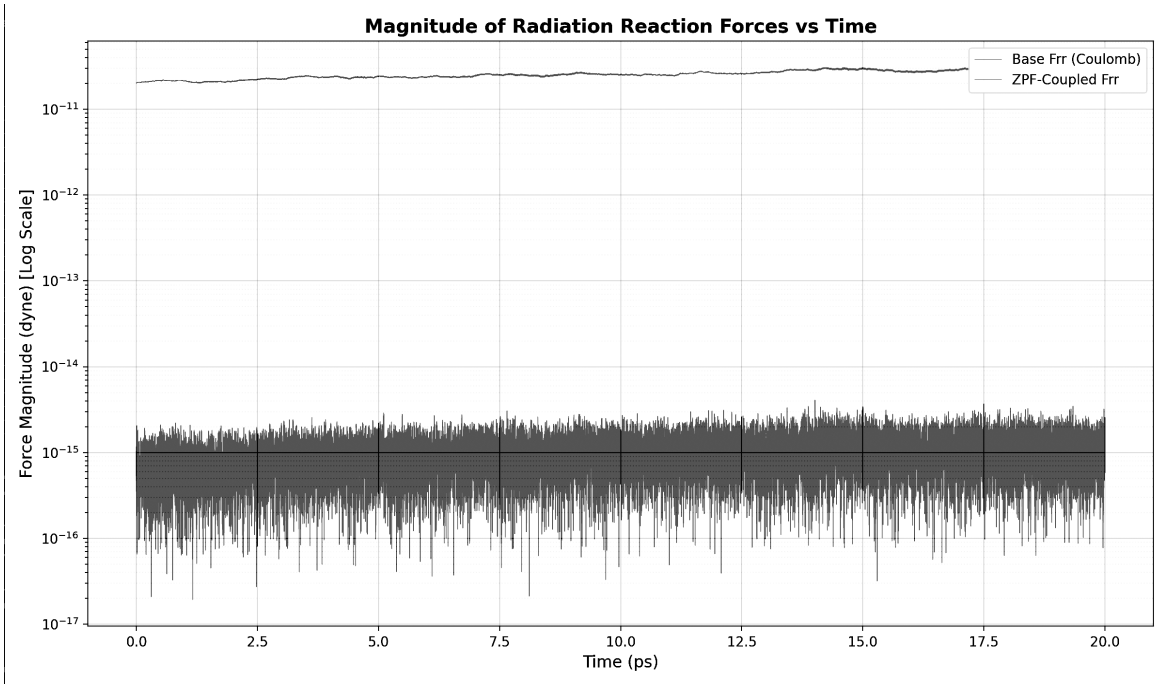


Fig. 1: The top curve is the magnitude of the radiation reaction due only to the Coulombic force. The bottom curve is the magnitude of the ZP Lorentz force part of the radiation reaction.

5.5. Tests on radiation reaction and dipole approximation using the SHO

The 3D SHO model, with a resonant frequency of $\omega_0 = 4.5 \times 10^{16}\text{s}^{-1}$, was used to further test the effects of the stochastic zero-point Lorentz force. During a 0.5 ps run, $r\,(t)$ varied from approximately 3×10^{-10}cm to 1×10^{-11}cm, due to the fluctuations induced by the stochastic ZP Lorentz force. Over this period of 0.5 ps, the variance $\langle r^2 \rangle$ with the ZP radiation included in $\vec{F}_{\text{rr}}$ minus the case of $\vec{F}_{\text{rr}}$ **without** the ZP radiation included was of the order of $1.5 \times 10^{-23}\text{cm}^2$, reflecting the small contribution of including the ZP Lorentz force in $\vec{F}_{\text{rr}}$, as shown in the previous subsection.

However, what was even more interesting was the following test. The same SHO simulation was run, but now with $\vec{F}_{\text{rr}}$ due only to the Coulombic force, and now the evaluation of $\langle r^2 \rangle$ with and without the dipole approximation being made. The difference between these two cases over the same 0.5 ps time period, was approximately $1.5 \times 10^{-26}\text{cm}^2$, about three orders of magnitude smaller than the previous test of including the ZP Lorentz force in $\vec{F}_{\text{rr}}$.

Thus, according to these tests, both situations of (1) making the dipole approximation and (2) not including the ZP Lorentz force in $\vec{F}_{\text{rr}}$, were each very small effects and made very good approximations. However, #1 of making the dipole approximation had an even smaller effect than #2 not including the ZP Lorentz force in $\vec{F}_{\text{rr}}$, significantly so, by many orders of magnitude, at least for the SHO conditions described here.

6.0. OVERVIEW OF SOFTWARE, CODE DETAILS, AND SIMULATION EXAMPLES

6.1. Software overview

The simulation program is written in C++ and compiled using Xcode. Program execution and input variables are controlled entirely through a config.json (JavaScript Object Notation) file, enabling users to adjust simulation conditions without recompiling the source code. There are approximately 25 input parameters available to dictate the simulation's behavior. These include seven parameters governing the adaptive Runge-Kutta integrator, the initial radial distance of the electron's orbit, toggles to enable or disable the ZP fields and/or applied EM fields (such as laser pulses), and the data output frequency for the integration points. Data visualization and statistical analysis are handled via post-processing Python scripts. For example, tracking how the electron's orbit evolves over time—specifically its semimajor axis a and eccentricity ε—is a primary focus. The scripts construct a histogram representing the duration the electron spends at each radial distance from the nucleus, which is then directly compared to the analytical QM probability distribution of Eq. (1). As described in the following subsections, Figures 2–4, which depict these various physical quantities, were generated entirely using this Python post-processing suite.

6.2. Several simulation outputs

The following figures illustrate the capabilities and stability of the numerical architecture developed in this work. While longer integration times will be employed in future physics-focused studies, the present 25 ps ensemble runs

clearly demonstrate the direction and potential of this work. All presented results strictly leverage the computational advancements described in Sections 3.0 and 4.0.

Figure 2 illustrates the time evolution of a classical electron orbiting the nucleus (proton) in the presence of electromagnetic ZP radiation over a 25 ps interval. The electron was initialized in a circular orbit of radius $r = 0.53\,\text{Å}$. The top panel displays the orbital radius $r\,(t)$ as a function of time, while the second panel tracks the semi-major axis $a\,(t)$. These top two plots appear nearly identical because the eccentricity $\varepsilon\,(t)$ (third panel) remains remarkably small throughout the simulation ($\varepsilon = 0$ corresponds to a circular orbit). The fourth panel records the polar angle of the normal to the orbital plane in radians; this angle remains below approximately 43 degrees (0.75 rad). Thus, the orbital plane changes only slowly across this run of nearly 190,000 orbital revolutions.

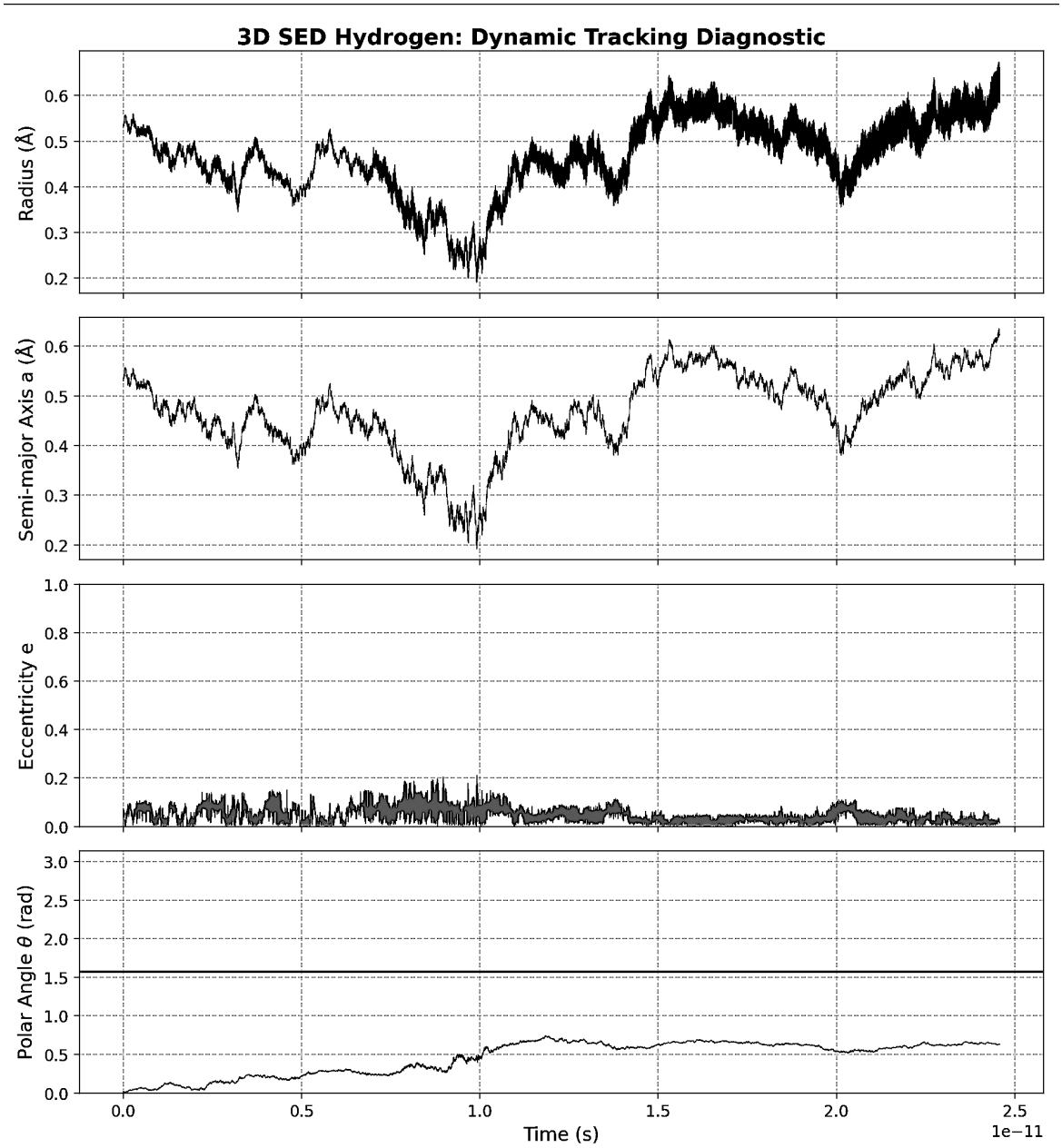


Fig. 2: Tracking diagnostic of the simulated electron's trajectory. The top plot shows the radial distance r, the distance from the radius of the nucleus as a function of time t. The second, third, and fourth panels display the semimajor axis a, the eccentricity ε, and the polar angle of the orbital plane's normal vector versus time, respectively.

Figure 3 takes the data of $r\,(t)$ vs. t from 17 independent 25 ps simulation runs to determine the likelihood of finding the electron at each radial position. Obtaining this histogram for a single electron requires tracking the trajectory line crossings at each radial value (effectively taking horizontal slices of the top plot in Fig. 2) and adjusting for the time spent in each bin via $|dr/dt|$. Clearly, Fig. 2 helps to illustrate how Fig. 3 arises from a classical physics diffusion-like process driven by the stochastic interaction with ZP radiation.

Figure 3 is likely the plot of most interest from our simulator, as it directly compares the probability density $P\,(r)$ of finding the classical electron at a distance r from the nucleus to the analytical prediction of QM (solid line), Eq. (1). Both the net area under the simulation histogram and the curve of Eq. (1) are normalized to unity. Crucially, no atomic ionization events occurred during any of the 17 simulation runs. The histogram shows strong future promise for even tighter agreement with Eq. (1) as integration times are extended beyond 25 ps, since longer durations will help "erase" the initial $r = 0.53\,\text{Å}$ memory of the ensemble, allowing further spatial spread to the left and right of the present distribution. Satisfyingly, the picosecond time domain required here to obtain meaningful 3D results agrees in order of magnitude with the time span from our 2003 2D simulations in [1].

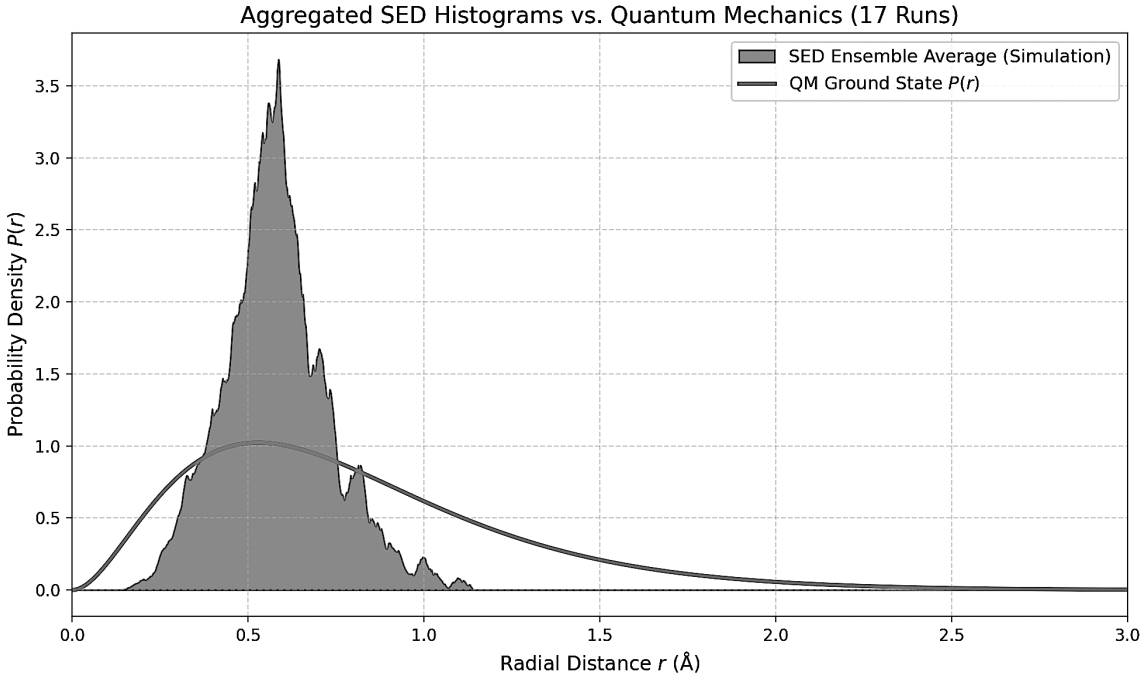


Fig. 3: Histogram of percent of time an electron spent at each radius, over the net of 17 electron simulation runs, overlaid with the QM ground state probability density function.

Further insight is shown in Fig. 4. Here the energy $E = \frac{m\vec{v}^2}{2} - \frac{e^2}{r}$ is plotted versus time in the top panel. The second panel displays the magnitude $\left|\vec{L}\right|$ of the angular momentum $\vec{L} = \vec{z} \times m\vec{v}$ of the electron in its orbit, while the

third panel shows L_z, the projection of $\vec{L}$ onto the z axis. Since the orbits are slowly changing ellipses, due to the strong Coulombic attraction, $E \approx -\frac{e^2}{2a}$, where a corresponds to the data tracked in the second panel of Fig. 2.

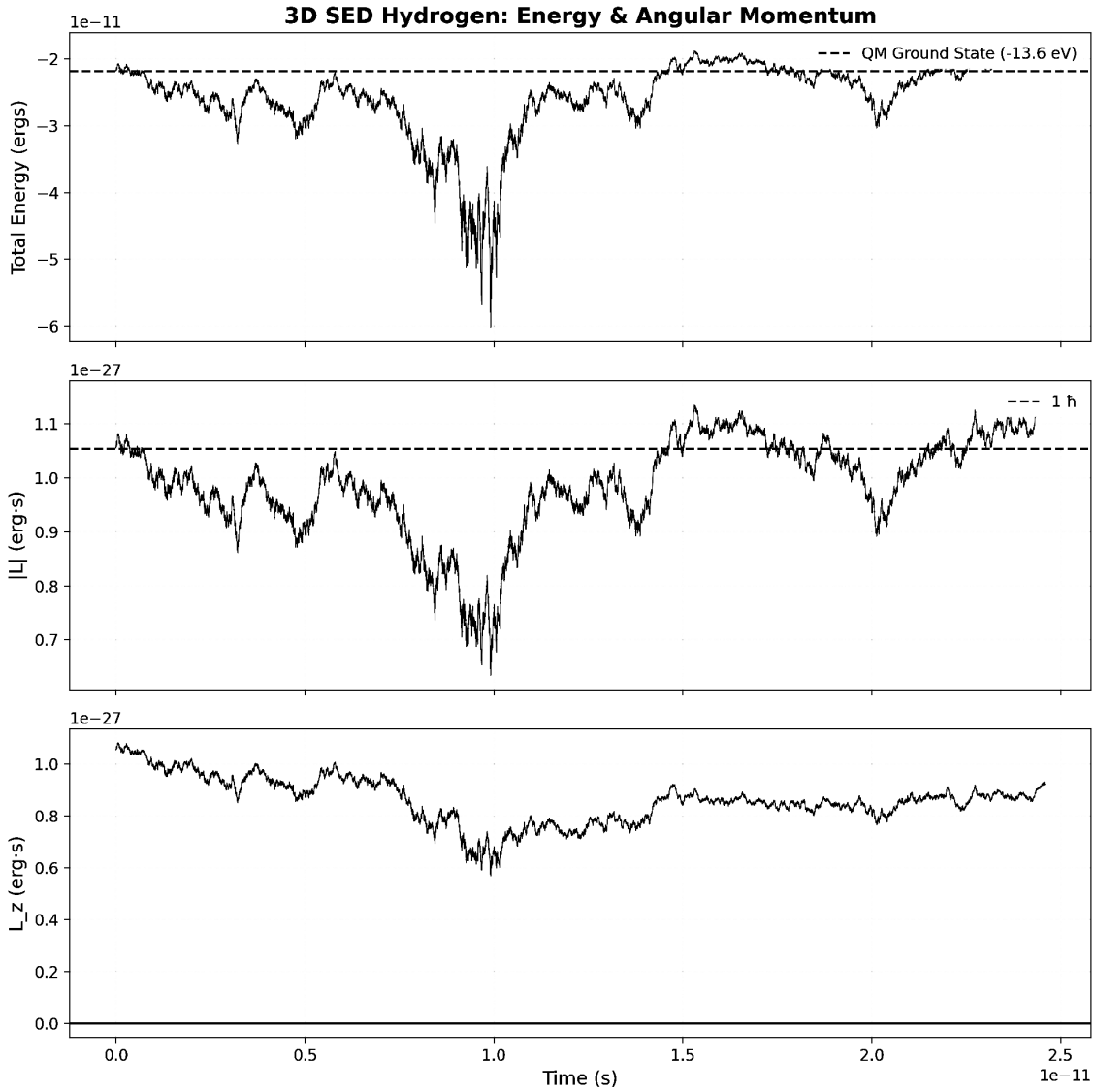


Fig. 4: Three panels of additional properties of the electron's "trajectory" (r vs. t) from Fig. 2. From top to bottom, respectively, is displayed the energy of the electron, the magnitude of the angular momentum, and the z-component of the angular momentum, all plotted versus t.

6.3. Software used to investigate artificial electron decays

The use of this software has already been shown in several ways so far, such as in several studies in Sect. 5.0, and in the figures produced in this section. While the analytical tests presented in Sect. 5.0 required specialized modifications to the C++ core during the development phase, all results and studies detailed in Sect. 6.0 can be reproduced directly using the publicly released GitHub code. Here is an example of such use and study.

The following convergence study serves as an example of leveraging the released software to investigate complex

physical phenomena. Constructing the ensemble for Fig. 3 provided insight into simulated electron behaviors - specifically, non-physical orbital collapses. In the simulation of Fig. 3, all electrons were initialized with a circular radius of 0.53 Å. While most orbits fluctuated and evolved naturally, occasionally an electron's radius would decrease dramatically and eventually "decay" into the nucleus. This behavior is expected in a classical system lacking zero-point (ZP) radiation, but it contradicts the physical mechanics of SED.

The theoretical expectation was outlined by Boyer in 1975 [10], who predicted that for small radii and corresponding high orbital frequencies, the electron resonates with the high-frequency regime of the ZP spectrum, which scales as ω^3. On average, this intense radiation resonance should overcome the classical radiation reaction at that small radius, driving the orbit to increase. Conversely, for larger radii and lower orbital frequencies, the electron resonates with the lower-frequency ZP spectrum and should be dominated by radiation reaction, causing the orbit to decay. Because the process is stochastic, a small orbit may probabilistically increase or decrease, but the probability of continuous decay approaches zero since the available ZP energy (ω^3) increases rapidly as the orbit shrinks. While this explanation has been simplified here in terms of circular orbits, simulations to date indicate that the eccentricity $\varepsilon\,(t)$ remains close to zero, meaning $r(t)$ is approximately the semimajor axis length, the more appropriate parameter.

To investigate this, a convergence study was conducted (Table 3). As the number of frequency bins (N) representing the ZP field in Eqs. (32) and (42) is increased, the numerical resolution improves, and the number of artificial decays decreases. For this study, an ensemble of 18 electrons was initialized in a tight circular orbit of 0.12 Å, representing a highly demanding computational condition compared to the 0.53 Å start in Fig. 2. A "decay" was triggered if the electron collapsed to a threshold of 0.0005 Å. The simulation tracked the ensemble until the first surviving electron reached 1.2 ps. in simulation time.

N (bins)	**# Decays**	**# Survivors**
1M	8	10
5M	7	11
10M	5	13
50M	3	15
100M	3	15

Table 3: Number of electron decays vs. number of frequency bins (N).

As demonstrated, the number of decays systematically decreases as the grid density increases. However, the execution time scales linearly with the number of bins. The bulk of the computational overhead involves summing over the frequencies within the sliding window (±5% of the central frequency) for the Lorentz force term at every time step. Consequently, running a simulation at $N = 100M$ takes approximately 100 times longer than $N = 1M$, making full ensemble studies like Fig. 3 highly impractical at maximum resolution without access to advanced supercomputing resources.

While Table 3 does not definitively prove that ZP radiation completely eliminates all classical decays, it is a clear indication that the observed collapses are numerical artifacts caused by the difficulty of adequately representing continuous high-frequency spectra in discrete simulations. For larger starting radii, such as 0.53 Å, the decay probability

per time unit decreases dramatically, though the probability of eventual decay inherently increases over long durations when N is made relatively small. Ultimately, these simulations serve not just to generate distributions, but as an analytical tool to uncover the subtle physics governing classical electron trajectories.

7.0. CONCLUDING REMARKS

Stochastic electrodynamics is a classical theory grounded in a simple but powerful premise: the necessary equilibrium between classical charged particles and classical electromagnetic radiation (see Ref. [6]). This dynamic equilibrium necessitates the existence of classical electromagnetic ZP radiation—a Lorentz-invariant field at temperature $T = 0$ [26], [27] that results in no heat flow during reversible, isothermal thermodynamic operations [28], [29]. Fundamentally, this ZP radiation must establish a stable equilibrium for charged particles in an open environment without containing walls.

Investigating this equilibrium, however, requires advancing beyond the more conventional, but already nontrivial, treatment of nonlinear oscillator binding potentials [40], [41], [42] to focus strictly on truly realistic atomic systems, such as the classical hydrogen atom [43]. This is precisely where the numerical and computational approaches discussed in this article become essential. Unlike QM and QED, addressing the physics of this simplest atomic system requires complex simulation capability. While this may initially seem counter-intuitive, it mirrors a well-established pattern in physics: when one foundational theory serves to derive another macroscopic or formalized theory—much like statistical mechanics deriving thermodynamics, or the Langevin microscopic equation deriving macroscopic diffusion—the underlying framework invariably involves significantly greater mathematical complexity. This is the exact relationship between the hydrogen atom in SED versus QM. Despite its conceptual elegance, the classical hydrogen atom in SED is fundamentally intractable without the use of advanced numerical computation.

This article has outlined the specific mathematical and computational architecture required to bridge that gap. We have described the symmetrical consideration of the pseudo-wave formulation, the large-scale indexing capability of logarithmic frequency gridding, and the handling of resonant behaviors between radiation and charged particle motion. These domain-specific advancements operate in tandem with the more recognizable advanced numerical integration methods required for solving highly nonlinear differential equations. All of these concepts are described in some detail here. They will likely be necessary not just for hydrogen, but also for more complicated systems such as helium and lithium, where multi-electron atomic systems are involved. Other important considerations involve externally applied EM fields such as from lasers or strong electric and magnetic field sources. Scaling this methodology to these more complex systems will likely need to build upon the work described here.

Data availability

The implementation of our method is available on GitHub https://github.com/mcallsen/BayLMD.

Need to submit software and get url to go here.

During the preparation of this work, the author used the program Gemini (Google AI) for aid in computer code generation. After using this AI program, the author reviewed and edited the work as needed and takes full responsibility for the content of this research.

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